Overvalued Geographic Familiarity in International Equity Management

Kevin Zheyuan Cui* Yinan Qiu†

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Abstract

Foreign education may provide country-specific expertise, but its value across international portfolios is less clear. Within funds over time and across similar funds, greater foreign-educated representation is associated with lower factor-adjusted net returns. High-conviction holdings in managers' education countries earn high alphas, but contribute little to portfolio alpha at actual weights. Outside these countries, foreign-educated teams select stocks less successfully in markets bordering or more socially connected to their education countries, where familiarity is expected to help. We connect these investment patterns to how funds recruit and evaluate managers. Funds are more likely to hire managers educated in a country after their holdings there underperform. These appointments are followed by short-lived local improvements. More broadly, foreign-educated recruits bring weaker prior team investment records than U.S.-educated recruits, and foreign-educated managers' retention is less sensitive to deteriorating fund performance than that of U.S.-educated colleagues. Together, the findings suggest that funds credit geographic familiarity more broadly than demonstrated investment performance warrants.

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*Department of Economics, Princeton University.
†Guanghua School of Management, Peking University.

1 Introduction

For international equity funds, selecting portfolio managers is itself an important investment decision. Information asymmetries make foreign firms harder for outside investors to evaluate (Kang and Stulz, 1997; Brennan and Cao, 1997). Managers educated abroad may help overcome these barriers by bringing knowledge of local firms and institutions. Consistent with a local informational advantage, Jagannathan et al. (2022) document superior stock selection in the home countries of international fund managers. Yet funds in our sample invest in about 19 countries on average with teams of roughly three managers. Country-specific expertise can improve part of that portfolio without extending to the many other markets the team manages. Fund returns provide noisy signals of investment skill (Fama and French, 2010), while team-based decisions make individual contributions difficult to assess. Funds may consequently rely on observable educational backgrounds, potentially crediting local expertise as evidence of broader investment ability. We ask how far geographic familiarity improves investment performance and whether funds distinguish its local value from broader investment ability when selecting and evaluating managers.

We find that greater foreign-educated representation accompanies lower factor-adjusted net returns. Within managers' education countries, selected high-conviction stocks earn high alphas, but their contribution at funds' actual portfolio weights is small. Beyond those countries, familiarity accompanies weaker selection precisely where it would be expected to help: foreign-educated teams select stocks less successfully in markets bordering or more socially connected to their education countries. This contrast between concentrated local value and poorer selection in related markets leads us to ask how funds use geographic familiarity when selecting and evaluating managers. Funds are more likely to hire managers educated in a country after their holdings there underperform, and these appointments are followed by short-lived local improvements. In the broader set of experienced recruits, however, foreign-educated managers bring weaker prior team records than U.S.-educated recruits, even in their potential country specialty. After appointment, their retention is less

sensitive to deteriorating fund performance than that of U.S.-educated colleagues. Together, the findings suggest that funds credit geographic familiarity more broadly than demonstrated investment performance warrants.

We study 1,663 U.S. actively managed international equity funds over 2000–2024, linking managers' education and dated fund assignments to portfolio holdings and returns. Foreign-educated managers have at least one recorded degree outside the United States; the U.S.-educated comparison group has recorded degrees only in the United States. The funds operate within a common U.S. asset-management market but invest across many foreign countries. Among funds employing foreign-educated managers, teams include about two such managers on average and invest in 18 foreign countries, 17 of which lie outside the team's recorded education countries. This lets us judge local expertise against the team's whole investment task. The same data also show what track records recruits bring and how their careers respond to the fund's later performance.

We begin with overall fund performance. We compare each fund with itself as its team changes over time, and with funds that share its benchmark in the same month. After adjusting for factor exposures, a change from an entirely U.S.-educated to an entirely foreign-educated team is associated with net returns about 70 basis points a year lower. The gap is nearly identical in gross returns, so fees do not explain the shortfall.

We next turn to the countries where managers studied. Relative to their benchmarks, funds put about 1.1 percentage points more weight in a country when one of their current managers studied there, and this overweight rises when the team gains a manager educated there and falls when it loses the last one. Where managers studied shapes where funds invest. We assess stock picking by comparing the returns funds earn on holdings in the same country and quarter, adjusted for stock characteristics. Foreign-educated teams show no clear average advantage over U.S.-educated teams within their education countries: the estimated difference is 0.2 basis points a year and statistically insignificant.

The strongest evidence of a local advantage comes from high-conviction positions. The positions that funds overweight most heavily within an education country earn a 4.0% annual alpha, 3.6 percentage points more than other positions in the same countries and 3.9 points more than similarly overweighted positions outside education countries, with every portfolio reweighted to the same country mix. These positions make up only 3.6% of funds' observed equity holdings, however, so even a 4.0% alpha would add only about 14 basis points a year to portfolio returns; estimated at the country weights funds actually hold, the contribution is close to zero. Familiarity thus has value in selected local investments, but that value contributes little to the overall portfolio.

Having located a concentrated local advantage, we ask whether it extends to markets close to the countries where managers studied. If studying in a country conveys useful information about related markets, foreign-educated teams should pick stocks better there. We measure closeness in three ways, a shared border, a common official language and social connectedness based on Facebook friendship links (Bailey et al., 2018), and compare how stock selection changes with closeness to where managers studied for foreign- and U.S.-educated teams. By all three measures, familiarity is worth far less to foreign-educated teams. Per dollar invested in those markets, a shared border is worth 484 basis points a year less to them than to U.S.-educated teams, a common official language 231 basis points less, and a one-standard-deviation increase in social connectedness 117 basis points less. For borders and social ties, foreign-educated teams also do worse in absolute terms: they select stocks 184 basis points a year worse in markets bordering an education country than in other outside markets, and their selection worsens as social connectedness rises. Closeness to an education country thus accompanies weaker stock picking precisely where familiarity should help, consistent with familiarity bias (Huberman, 2001; Pool et al., 2012).

We next turn from investment outcomes to the demand for country expertise. Weak returns in a country's market and weak performance of the fund's own holdings there each predict the appointment of a manager educated in that country. Relative to comparable

funds that do not hire, the hiring funds then select stocks better in that country, with no statistically clear improvement in their holdings outside education countries or in overall NAV. The local gain is also short-lived. Following the same appointments and comparison funds for two years, the improvement is 195 annualized basis points in the first year and near zero in the second, a decline of 205 basis points. Funds thus have a local reason to make these appointments, but the payoff is concentrated in the first year.

The local payoff explains why funds seek country knowledge, but not what track records they accept from the managers who supply it. Using the broader set of experienced appointees, we compare foreign- and U.S.-educated recruits appointed by the same fund on their previous teams' stock-selection performance over the four quarters before appointment. Foreign-educated recruits' prior teams underperform by 86 basis points a year, after accounting for previous funds' benchmarks, school rankings and other observed qualifications. Their previous funds also attracted weaker investor flows. Funds thus accept weaker investment records from foreign-educated recruits, and a stronger fundraising record does not make up the difference.

The general record averages performance across countries, so it could hide strong performance where a recruit's local knowledge should matter most. We therefore examine markets the hiring fund already holds but in which no continuing manager studied. In each such market, the previous teams of recruits educated there earned adjusted returns 274 basis points a year below those of U.S.-educated recruits appointed by the same fund, controlling for prior overall fund returns. The weaker records extend to the very markets where the recruits' education should give them an advantage.

After appointment, we ask whether funds give realized performance the same weight when evaluating foreign- and U.S.-educated managers. Comparing colleagues on the same fund, we find greater relative retention for foreign-educated managers after poor returns and smaller relative gains in new assignments after good returns. A ten-percentage-point deterioration in benchmark-relative quarterly returns widens the foreign-minus-U.S. retention difference

by 1.45 percentage points. A ten-point improvement lowers their relative probability of receiving a new fund assignment by 1.07 points. These paired responses suggest that funds place less weight on realized performance when evaluating foreign-educated managers.

Finally, we consider whether geographic familiarity offers advisers a commercial payoff through greater asset gathering or higher fees. We find no statistically clear association with greater subsequent flows or higher reported fee rates. The measured commercial outcomes do not establish a compensating benefit for advisers.

The evidence separates a real but narrow local advantage from broader investment ability. Funds act on the local advantage: they hire for it after country losses, and local stock selection improves for about a year, without a clear gain for the portfolio as a whole. Yet the records they accept at entry are weaker, including in the recruits' potential specialty, and later career outcomes respond as if realized performance received less weight. Together with lower factor-adjusted fund returns and weaker stock picking in familiar outside markets, these patterns suggest that funds credit geographic familiarity more broadly than demonstrated investment performance warrants. The interpretation resembles the "kernel of truth" in Bordalo et al. (2016), where beliefs exaggerate real differences between groups. Here the real difference is local expertise, and the exaggeration is to treat it as evidence of broader investment ability.

The paper contributes to research on the scope of investment information. Geographic, educational and offshore connections can improve investment decisions (Coval and Moskowitz, 2001; Cohen et al., 2008; Baik et al., 2010; Bai et al., 2022), while familiarity and perceived competence can affect portfolios without a corresponding advantage (Heath and Tversky, 1991; Huberman, 2001; Grinblatt and Keloharju, 2001; Graham et al., 2009; Pool et al., 2012). Relatedness also shapes international allocations (Schumacher, 2018; Kuchler et al., 2022). We separate the concentrated value of local stock choices from the poorer selection that accompanies familiarity beyond education countries, and relate that distinction to how funds select and evaluate managers. The concentration of value

complements evidence on industry specialization, active share and best ideas (Kacperczyk et al., 2005; Cremers and Petajisto, 2009; Antón et al., 2021).

The personnel evidence contributes to research on manager selection and evaluation. Education and background predict fund outcomes (Chevalier and Ellison, 1999a; Gottesman and Morey, 2006; Chuprinin and Sosyura, 2018), and performance matters for replacement and reassignment (Khorana, 1996; Chevalier and Ellison, 1999b; Fang et al., 2014). Qualifications provide signals of productivity (Spence, 1973; Phelps, 1972; Aigner and Cain, 1977); subsequent outcomes allow employers to update those assessments (Farber and Gibbons, 1996; Altonji and Pierret, 2001), although fund returns are noisy (Fama and French, 2010). We show that funds accept weaker general and country-specific records from foreign-educated recruits, and that these managers' careers respond less to realized fund performance than those of U.S.-educated colleagues. The result complements work on professionals' beliefs (Linnainmaa et al., 2021) by examining personnel and portfolio outcomes surrounding a particular qualification.

Our closest comparison is Jagannathan et al. (2022), who document a home-field investment advantage in international mutual funds. We ask whether funds credit that background beyond the markets where it helps, by connecting performance across the whole portfolio to how funds recruit and evaluate the managers who have it.

Section 2 describes the institutional setting, the data and measures, and the prevalence of foreign-educated managers. Section 3 proceeds from fund NAV to education-country allocation and returns, then to familiarity in outside countries. Section 4 examines country-linked hiring and its local payoff, accepted recruitment records, later career responses and commercial outcomes. Section 5 concludes.

2 Setting, Data and Measures

2.1 Institutional setting

International equity funds give U.S. investors access to equity markets abroad. Morningstar classifies a fund in this group when foreign holdings account for at least 40% of its equity portfolio on average over the preceding three years. In a typical sample quarter, about 760 active funds operate, with mean assets of $1.6 billion. Most have global or global-ex-U.S. benchmarks; others follow emerging-market, regional or single-country mandates. Funds invest in 19.0 countries on average and hold a median of roughly 80 stocks, compared with more than 5,000 stocks in about 50 countries in the Morningstar Global ex-US index. This setting lets us distinguish the allocation across countries from the selection of stocks within each country, while observing the overall return investors receive.

Investment decisions are delegated to small teams. The average fund has 3.1 managers with mean tenure of 5.6 years, and managers often serve on several funds simultaneously. Among funds with at least one foreign-educated manager, the average team has 4.1 managers, including 2.0 foreign-educated managers, and holds stocks in 18.4 foreign countries. Of these markets, 17.2 lie outside all recorded education countries of the current team. The team's mandate therefore extends well beyond the countries linked to its members' education. Because returns and holdings are observed only for the team, even when its members divide research responsibilities, the performance tests use team education and the appointment tests use the records of a recruit's previous teams. Individual spells identify later departures and assignments, and a manager's education countries mark that manager's potential specialty.

The adviser or sub-adviser selects the managers assigned to a fund. Appointees may come from another firm, from the same family or from outside the recorded fund-management population. Experienced recruits bring observable fund-team records alongside the educational qualifications in their biographies. These records show what performance funds accept at

appointment, and subsequent spells show how colleagues' careers vary with shared fund performance: the first comparison describes entry, the second later evaluation.

2.2 Funds, holdings and returns

Our sample consists of U.S. actively managed international equity funds in Morningstar over 2000–2024, including funds that later liquidated or merged.[1] For each fund we observe quarterly stock holdings, the holdings of its assigned benchmark, monthly net returns and assets in U.S. dollars, reported gross returns, and annual expenses and turnover. Security returns and market capitalizations come from Datastream. We assign each company to the country of its primary business activity rather than its listing, which aligns country weights with economic exposure (Demirci et al., 2022).[2]

2.3 Manager education

Morningstar biographies list each manager's post-secondary institutions, and dated fund spells identify the managers serving together at each quarter-end, as well as appointments and departures. We map institutions to countries by hand and exclude professional certificates and executive programs.[3] A manager is foreign-educated if at least one recorded degree is from outside the United States and U.S.-educated if all recorded degrees are from the United States. Education is known for 85.5% of manager-quarters in the NAV sample.

Our main variable, ForeignShare_{jq}, is the fraction of fund j's managers at quarter-end q who are foreign-educated, among those with known education. The fraction with unknown

[1] The extraction sets U.S. Category Group to International Equity, Oldest Share Class to Yes and Index Fund to No, without restricting the sample to surviving funds. Each fund is identified by the ISIN of its oldest share class, whose returns we use; identifiers that share a Morningstar fund identifier form one economic fund, the unit of clustering.

[2] Where total-return series are unavailable, returns are constructed from prices and dividends. Missing returns are excluded rather than set to zero.

[3] A manager with degrees from several countries is linked to each. Conflicting profiles are coded as unknown education. Education is reconstructed from collected profiles; it is not an archive of the biography visible to an employer at the appointment date.

education, UnknownShare_{jq}, enters separately.[4] Coefficients compare an entirely foreign-educated team with an entirely U.S.-educated team; a ten-percentage-point change is one tenth as large. The measure captures where managers studied, not their nationality or the diversity of the team (cf. Lu et al., 2024). Appointment and career tests use the individual indicator $\mathbb{1}_i^{\text{Foreign}}$, which equals one if manager i is foreign-educated.

For a given team, education countries are the non-U.S. countries in which at least one current manager studied, and we refer to holdings there as local. All other foreign markets are outside countries. In the country panel, $\mathbb{1}_{jcq}^{\text{Link}} = 1$ if a current manager studied in country c. Because a team can include foreign-educated managers without a link to a particular country, $\mathbb{1}^{\text{Link}}$ and ForeignShare capture different things. We measure how closely a market is related to an education country by shared borders and official language from CEPII GeoDist (Mayer and Zignago, 2011) and by the Facebook Social Connectedness Index (Bailey et al., 2018). For U.S.-educated managers, these ties are measured relative to the United States.

Except where noted, the regressions control for five observed team traits, covering gender composition, CFA charters, biography coverage and fund-management experience, to account for differences in team composition and experience.[5]

2.4 Performance measures

Our first measure is the fund's net return, which captures the outcome for investors across the entire portfolio, including country allocation, stock selection, cash, trading and costs (Kacperczyk et al., 2008). To compare funds with different mandates and styles, we deduct the return explained by predetermined exposures to thirteen factors: the five factors and

[4] For a team with two foreign-educated, one U.S.-educated and one unknown-education manager, the foreign-educated share is 2/3 and the unknown-education share is 1/4.

[5] The traits are the female share among managers with known gender, the CFA share among managers with biographies, the fractions with unknown gender and no biography, and mean log one plus years since the manager's first recorded fund spell. Appointment tests also include indicators for degrees from top-100 and 101–500 institutions in the 2003 Academic Ranking of World Universities, so ranking-controlled appointment comparisons begin in 2004.

momentum for developed ex-U.S. markets and for emerging markets (Fama and French, 2015), plus the U.S. market. For fund j in month m,

$$r^a_{jm} = R^{\text{net}}_{jm} - R^f_m - \hat{\beta}'_{j,m}\mathbf{f}_m. \tag{1}$$

Here R^{net}_{jm} is the net return, R^f_m the one-month Treasury-bill return, $\mathbf{f}_m$ the factor returns and $\hat{\beta}_{j,m}$ the fund's loadings estimated over the preceding 60 months. Estimating loadings before the evaluated month lets exposures evolve and keeps the evaluated return out of its own loading estimate.[6] We express this factor-adjusted return in annualized basis points, $R^a_{jm} = 12 \times 10^4 r^a_{jm}$.

Our second measure evaluates stock selection within a country. A fund can benefit from a strong national market without picking better companies there; comparing its holdings with similar local stocks separates the choice of stocks from the choice of countries (Brinson and Fachler, 1985; Coval and Moskowitz, 2001). For each country c in which fund j holds stocks at quarter-end q, we compare the return on the fund's holdings in that country over the next three months with the return on a matched portfolio of similar local stocks, following characteristic-adjusted performance evaluation (Daniel et al., 1997; Wermers, 2000). Each stock k the fund holds receives a benchmark return. Within the country and quarter, stocks are sorted into terciles by market capitalization at the quarter-end and, within each size tercile, into terciles by their cumulative return over the eleven months before the quarter-end month. The benchmark return $\tilde{r}_{km}$ in month m is the capitalization-weighted return of the other stocks in the same size and past-return group, excluding stock k itself.[7] In each of

[6] Loadings require at least 36 observed returns. Without that history, we use loadings estimated on other funds with the same benchmark over the same window, excluding the focal fund, with broader peer groups where needed; 32,584 of the 206,847 main-sample months use peer loadings. The adjustment deducts only the factor component of the loading regression.

[7] The comparison stocks are all stocks held by sample funds in that country and quarter, and their capitalization weights drift with returns within the quarter. When fewer than five other stocks in the group have a return that month, the benchmark uses the size tercile alone and, failing that, the country's market return; about three quarters of holdings capital receives the full size and past-return match.

the three months m that follow quarter-end q, the return on the fund's holdings in country c in excess of these benchmarks is

$$r_{jcm}^{\text{adj}} = \frac{\sum_k \omega_{jkm} \left(r_{km} - \tilde{r}_{km}\right)}{\sum_k \omega_{jkm}}, \tag{2}$$

where r_{km} is stock k's return in month m, and ω_{jkm} is stock k's weight in the fund's portfolio at quarter-end q, grown with the stock's returns from the quarter-end through month $m-1$. The sums run over the fund's stocks in country c that have a return in month m; stocks with missing returns are dropped rather than set to zero. We refer to the fund's holdings in a country as its country basket. The fund's adjusted country return, R_{jcq}, is the average of r_{jcm}^{adj} over the three months, annualized and expressed in basis points per dollar of the basket. A positive value means that the fund's holdings in country c beat similar local stocks. Holdings and teams observed at quarter-end q are thus paired with returns over the following quarter, and both are updated each quarter. The same measure underlies the prior-record tests in Section 4.

NAV coefficients describe performance per dollar of fund capital; country-return coefficients describe performance per dollar invested in a particular market. The country tests locate geographic stock-selection patterns within the broader NAV comparison. Because the regressions use different comparisons and weights, their coefficients are not components of the NAV coefficient. Appendix Table A20 summarizes the outcomes, units and timing.

2.5 Sample and summary statistics

All analyses start from the same 1,663 funds with manager records, an assigned benchmark and valid country weights, and each test uses the observations its outcome requires. The main NAV sample has 206,847 fund-months at 1,418 funds with at least 24 months of eligible return history; including all 1,536 funds with usable returns gives a similar coefficient. The joint country-selection panel has 974,924 fund-country-quarters at 1,428 funds, including

38,806 in education countries; the outside-country panel used in Section 3.5 has 936,076 at 1,423 funds.

Table 1 reports summary statistics, including the fund, team and country counts described above; annual expenses average 1.22%. The sample funds' combined net assets averaged $1.3 trillion a year, rising from $497 billion in 2000 to $2.2 trillion in 2024 and peaking at $2.5 trillion in 2021.[8] The mean foreign-educated share is 0.35, and about half of fund-quarters have no foreign-educated manager. Panel E reports the country and manager counts on the same fund-quarters for teams with at least one foreign-educated manager. Among the 2,992 managers with recorded education, 26.9% studied only outside the United States, and the United Kingdom is the most common foreign education country. Figure 1 shows that the fraction of funds with at least one foreign-educated manager rose from 27.8% in 2000Q1 to 49.5% in 2024Q4.[9]

Panel standard errors are clustered by fund, allowing for dependence over time and across a fund's countries (Petersen, 2009; Cameron and Miller, 2015). Unless stated otherwise, fund-months and fund-quarters receive equal weight; precision-weighted estimates are supplementary checks (Solon et al., 2015). Portfolio alphas use monthly time series.[10]

3 Investment Performance and Geographic Familiarity

We begin with NAV returns to assess performance across the whole fund, including any local gains. We then examine allocation and returns in managers' education countries, where direct experience should be most useful. Finally, we ask whether familiarity also helps stock

[8] Combined assets sum the fund-year net assets of Panel B across funds, including all of each fund's share classes; 567 funds report assets in 2000 and 869 in 2024.

[9] Each fund receives equal weight, and funds with unknown education remain in the denominator; because the figure covers all active sample funds, its shares differ from the NAV-sample statistics in Table 1. Appendix A.1 examines the foreign-educated share of new appointments.

[10] Country-allocation regressions cluster by fund identifier. Portfolio regressions use a monthly risk-free rate converted from the Treasury-bill yield, conventional standard errors for individual alphas and robust Wald tests for differences between alphas.

selection in markets outside those countries. The NAV test evaluates the full mandate; the holdings tests distinguish concentrated local value from performance elsewhere.

3.1 Fund returns and expenses

The NAV comparison asks whether greater foreign-educated representation accompanies better performance across the entire fund, after accounting for market and style exposures (Carhart, 1997; Fama and French, 2010). The outcome R^a_{jm} is the factor-adjusted net return of Equation (1), in annualized basis points. We follow the same fund as its team changes and compare it with funds sharing its benchmark in the same month:

$$R^a_{jm} = \alpha_j + \lambda_{bm} + \rho \text{ForeignShare}_{j,q-1} + \Gamma' X_{j,q-1} + u_{jm}. \tag{3}$$

Here q is the quarter containing month m, so the team at the preceding quarter-end determines ForeignShare for the three months of quarter q and education precedes the evaluated returns. Fund effects α_j account for persistent differences across funds; benchmark-month effects λ_{bm}, where b is the fund's assigned benchmark, compare funds facing the same mandate and date. The coefficient ρ is the annualized return difference associated with a zero-to-one change in the foreign-educated team share.

Staffing can respond to the fund's recent performance and business conditions. We control for those conditions using prior-year benchmark-relative returns and volatility, preceding-quarter flows and lagged assets. The remaining controls capture team scale and observed qualifications, expenses, turnover and top-ten holdings concentration.[11] Because portfolio operations may themselves reflect earlier staffing decisions, we also report the model without

[11] X contains unknown education; log team size and log lagged assets with spline terms; prior-year relative performance and volatility; preceding-quarter flows; prior-year expenses and turnover; prior-quarter top-ten concentration; and the five team traits defined in Section 2. Prior-year returns require nine observed months and use benchmark peers excluding every share class of the focal fund. Volatility is log one plus annualized relative volatility; flows enter directly in percentage points of assets at the beginning of the preceding quarter, with a missing-flow indicator. Expenses are annual basis points, turnover is log one plus the reported percentage divided by 100, and concentration is the observed top-ten equity weight. Missing-value indicators retain unavailable controls.

expenses, turnover and concentration. Fund-months receive equal weight, with standard errors clustered by economic fund.

Funds earn lower factor-adjusted net returns when a larger share of their managers is foreign-educated. Table 2, column 1, estimates −69.5 annual basis points for the full zero-to-one difference in the team share, with a standard error of 35.4. At the sample mean share of 0.35, the fitted difference from an otherwise comparable zero-share observation is approximately 24 basis points a year. For scale, the full-share coefficient exceeds half the sample's average annual expense ratio of 1.22%. The sample and operating-control comparisons in Table 2 and Appendix Table A7 give estimates between −66 and −70 basis points. Weighting funds by the precision of their returns gives −70 with $p = 0.002$. The estimated magnitude is similar across these sample and weighting choices; the equally weighted model remains the primary estimate. Because NAV includes education-country investments, the estimate already incorporates any local gains.

To ask whether the shortfall arises before or after expenses, we replace net NAV with reported gross NAV in Equation (3), keeping exposures, observations, controls, effects and weights fixed. On 205,773 common fund-months, both coefficients are approximately −71 annual basis points (Table 3). Their paired difference, $\rho_{\text{gross}} - \rho_{\text{net}}$, is −0.47 basis points, with a 95% interval of $[-2.45, 1.51]$. This interval is narrow relative to the return gap, and the comparison without operating controls gives the same conclusion: the shortfall arises in gross returns, not in the expenses deducted from them.

The appendix examines three other readings of this gap. First, a poor match between managers' education countries and the fund's mandate does not explain it: holding the foreign-educated share and education countries fixed, returns are weaker, not stronger, where those countries carry more weight in the fund's benchmark (Appendix B.1). Second, funds do hire country specialists after local losses (Section 4.1), but such staffing responses do not produce the association: adding the team's composition four quarters ahead leaves it essentially unchanged (Appendix B.4). Third, the gap is not a transitional cost of new

appointments, because it is present among foreign-educated managers who have been at the fund for more than two years (Appendix B.5). The holdings tests next locate where geographic familiarity has investment value, beginning in managers' education countries, where direct experience gives the clearest reason to expect an advantage.

3.2 Allocation to education countries

We begin the geographic analysis by asking whether funds put more capital in a country when one of their managers studied there. Because a large national market receives a large weight mechanically, we measure each country's weight relative to the fund's assigned benchmark. For fund j, country c and month m, we estimate

$$\text{Overweight}_{jcm} = \alpha_j + \lambda_{cm} + \psi \mathbb{1}^{\text{Link}}_{jcm} + \Gamma' X_{jm} + e_{jcm}. \tag{4}$$

Here c ranges over non-U.S. countries, Overweight_{jcm} is the percentage of the fund's portfolio invested in country c minus that country's weight in the fund's assigned benchmark, in percentage points, and $\mathbb{1}^{\text{Link}}_{jcm}$ equals one if a current manager studied in c. The coefficient ψ measures how much more a fund overweights a country when one of its current managers studied there. A second specification replaces the indicator with LinkedShare_{jcm}, the fraction of the fund's current managers, including those with unknown education, who studied there, so that ψ compares an entirely linked team with an unlinked one. Country-month effects remove what all funds do in a country in a given month. We first include fund effects α_j and then replace them with fund-month effects α_{jm}; the latter compare countries within the same portfolio in the same month, so ψ then measures how a fund distributes its capital between linked and unlinked countries. Controls X contain the five team traits and are absorbed

by fund-month effects.[12] The allocation coefficient describes where funds place capital; the following return tests assess the stock choices made there.

Both measures show an active education-country tilt (Table 4). The indicator coefficients are 1.11 percentage points with fund effects and 1.09 with fund-month effects. The linked-share coefficients are 2.63 and 2.59, all significant at 1%. The latter compare a team with no linked manager with an entirely linked team; a ten-percentage-point increase in linked-manager share corresponds to about 0.26 percentage points more country overweight. The similar within-fund-month results show that the education-country tilt persists when comparing destinations within the same portfolio and month.

Figure 2 traces the overweight around changes in the team. An event is the arrival of the first manager educated in country c, or the departure of the last such manager. With $\text{Overweight}_{jcq} = w_{jcq} - w^b_{jcq}$, where w_{jcq} and w^b_{jcq} are the fund's and its benchmark's percentage weights in country c at quarter-end q, we estimate separately for arrivals and departures

$$\text{Overweight}_{jcq} = \alpha_{jc} + \lambda_{cq} + \mu_{jq} + \sum_{\ell \neq -1} \psi_\ell \, \mathbb{1}\{q - q^*_{jc} = \ell\} + e_{jcq}. \tag{5}$$

Here q^*_{jc} is the quarter of the event for fund j and country c, so $\mathbb{1}\{q - q^*_{jc} = \ell\}$ equals one ℓ quarters after the event, for ℓ from -6 to 12, and each ψ_ℓ measures the change in the country's overweight relative to the quarter before the event, $\ell = -1$, which is omitted. Fund-country effects remove the fund's usual weight in that country, country-quarter effects remove changes common to all funds investing there, and fund-quarter effects remove changes in the fund's overall portfolio.[13] Weights rise about 0.5–0.7 percentage points after arrivals

[12] The equally weighted panel starts with all 1,663 funds and contains approximately 5.6 million fund-country-months from January 1999 to December 2024; because these tests use only holdings and benchmark weights, not subsequent returns, they also include the 1999 holdings that precede the return sample; the fund-month models retain 1,661 funds after singleton fixed-effect groups are dropped. A missing-trait indicator retains teams without biographies, and the team record assigned to each month defines $\mathbb{1}^{\text{Link}}$ and X. Errors are clustered by fund identifier.

[13] The 977 arrivals and 793 departures use up to twelve-quarter windows, a six-quarter clean pre-period, no overlapping events and truncation if the link reverses. The panels retain fund-identifier clustering.

and fall 0.2–0.55 after departures, without a clear movement beforehand. Capital follows the team's educational geography as its membership changes. The next question is whether stock selection in these countries earns a return advantage over U.S.-educated teams.

3.3 Stock selection in education countries

Funds allocate more capital to managers' education countries. Table 5 asks whether they also select stocks better there, on average, than U.S.-educated teams investing in the same markets. This comparison evaluates the entire country basket before we examine high-conviction positions separately. We use the adjusted country return, R_{jcq}, defined in Section 2.4, in a joint panel of education and outside countries:

$$R_{jcq} = \alpha_{jc} + \lambda_{cq} + \beta \mathbb{1}^{\text{Link}}_{jcq} + \gamma \text{ForeignShare}_{jq} + \Gamma' X_{jcq} + \varepsilon_{jcq}. \tag{6}$$

Here $\mathbb{1}^{\text{Link}}_{jcq}$ equals one if a current manager studied in country c, and ForeignShare_{jq} is the team's foreign-educated share. Two coefficients describe distinct comparisons. The coefficient γ is the difference between an entirely foreign-educated team and an entirely U.S.-educated team in markets where no current manager studied. The coefficient β is the additional difference when a current manager studied in the market. The comparison of interest, a fully foreign-educated team investing in its education country versus a U.S.-educated team investing in the same country, is therefore $\beta+\gamma$. Fund-country effects compare the same fund in the same market over time, and country-quarter effects compare funds investing in the same market and quarter, so β is identified when a link to a market begins or ends within a fund. Controls X contain log country stock count, log team size, unknown-education share and the five team traits. Countries have equal weight within a fund-quarter, and standard errors are clustered by fund.

The joint panel contains 974,924 fund-country-quarters at 1,428 funds, including 38,806 in education countries; links begin or end in 1,037 fund-country pairs at 525 funds.[14]

The comparison of interest adds the two coefficients. In column 1 of Table 5, $\beta + \gamma$ is +0.21 annualized basket basis points (standard error 56.61): in their education countries, foreign-educated teams show no clear average advantage over U.S.-educated teams investing in the same markets. Neither coefficient is statistically significant on its own; they matter here only through their sum. The precision-weighted comparison in column 2 is similar (+5.57 basis points, standard error 51.39). This is an average across all holdings in those countries, distinct from whether a fund improves after a targeted appointment, which we examine in Section 4.1.

3.4 The concentrated local value of geographic expertise

An average country basket can conceal information concentrated in a few companies. We therefore compare the positions managers overweight most with their ordinary holdings under a common country mix, which isolates the choice of stocks, and then ask how much these positions add to the funds' returns.

The test asks whether the fund's largest active stock bets within an education country outperform its other holdings there. We identify those positions before observing their returns using a country-normalized version of the I-measure of Antón et al. (2021):

$$I_{jkq} = \frac{\omega_{jkq}}{\sum_{k' \in \Omega_{jcq}} \omega_{jk'q}} - \frac{\mathrm{MC}_{kq}}{\sum_{k' \in \Omega_{jcq}} \mathrm{MC}_{k'q}}. \tag{7}$$

Here k is a stock that fund j holds in country c at quarter-end q, Ω_{jcq} is the set of such stocks and k' runs over it, ω_{jkq} is the stock's weight in the fund's portfolio, as in Equation (2), and MC_{kq} is its market capitalization at quarter-end q. The measure subtracts the stock's

[14] The identifiers represent 1,429 oldest share classes. Formation quarters run from 1999Q4 through 2024Q3, with payoffs from January 2000 through December 2024. Country weights are renormalized over all supported foreign destinations within each fund-quarter. Unknown stock returns are excluded from covered baskets, not set to zero.

capitalization share among those holdings from its weight in the fund's country basket. A positive value identifies an active overweight within the local stock selection. Normalizing within countries prevents a large country allocation from mechanically making every stock there a high-conviction position. High conviction here refers to revealed active weights among stocks the fund holds.

Within each fund-country-quarter, the top I-measure quartile defines high conviction and the other three quartiles define ordinary holdings. We form calendar-time portfolios for each group, separately in education and outside countries. Each portfolio pools the group's positions across all funds: within a country, positions are weighted by their dollar value, and countries are weighted by their shares in the Morningstar Global ex-US index. Reweighting to these benchmark country shares removes differences in aggregate national-market weights from the group comparisons. Each portfolio's alpha comes from the calendar-time regression

$$R_{pm} - R_m^f = \alpha_p + \beta_p' \mathbf{f}_m^{(10)} + \beta_p^G (R_m^G - R_m^f) + \varepsilon_{pm}. \tag{8}$$

Each regression uses the 300 months from January 2000 to December 2024. The portfolio index p identifies the country group and weighting rule; all local and outside conviction portfolios use this same model. All regression returns are monthly decimals: R_{pm} is the constructed portfolio's return, $\mathbf{f}_m^{(10)}$ contains the developed- and emerging-market five factors, and R_m^G is the return on the Morningstar Global ex-US index, with loading β_p^G. This common portfolio benchmark differs from the fund-specific assigned benchmarks in the NAV model. The focal intercept α_p is monthly excess performance beyond these exposures; multiplying it and its standard error by 12×100 gives annual percentage points, or by 12×10^4 gives annual basis points. Direct alpha differences use stacked portfolio series and robust Wald tests on intercept equality. The intercept measures the performance of a strategy that holds the disclosed positions, not the funds' realized returns.

The highest active local bets have a substantial alpha advantage. The top education-country quartile earns a 4.0% annual alpha (standard error 1.3), compared with 0.4% for ordinary local holdings (Table 6). Their direct 3.6-percentage-point difference is significant at 1%. Because both portfolios are reweighted to common benchmark country shares, this comparison evaluates stock selection within the same local investment opportunity set. The return advantage is concentrated in the local stocks that receive the largest active weights.

Comparing top positions across country groups provides a second test of the local link. Top positions in education countries outperform top positions outside them by 3.9 percentage points a year ($p = 0.002$). With the 3.6-point local top-versus-ordinary difference, this result places the advantage in high-conviction education-country holdings. Figure 3 plots the four alphas, their confidence intervals and the direct tests. Appendix B.6 reports eight alternative constructions, including two alternative conviction measures, the exclusion of U.K. equities, undergraduate-only education links, other risk models and returns net of trading costs; the top-versus-ordinary difference remains significant at 5% in six of them (Appendix Table A10).

Figure 4 compounds the same monthly gross strategy returns over 2000–2024, starting with one dollar before January 2000. It shows when the local top portfolio separates from ordinary holdings and the benchmark, alongside the outside-country comparison. The curves track gross returns on the constructed strategies under benchmark country weights.

How much do these positions add to the funds' returns? They account for 3.6% of the sample funds' observed equity holdings. Even if every dollar in them earned the 4.0% annual alpha, they would add only about 14 basis points a year to portfolio returns ($0.036 \times 4.0\%$). The estimated contribution is smaller still and close to zero, because the same stocks earn much less at the country weights funds actually hold than under the benchmark country mix (Appendix B.7).

The local advantage is real but narrow: it lies in the few stocks managers overweight most and adds little to the portfolio as a whole. We next ask whether useful information extends to markets connected to managers' education countries.

3.5 Stock selection and familiarity outside education countries

The local results identify useful information in selected education-country investments. We return to characteristic-adjusted country returns to ask whether this advantage extends to markets connected to those countries.[15] If information acquired in one country extends to nearby or closely connected markets, foreign-educated teams should select stocks better in those markets. If familiarity instead creates confidence beyond the reach of useful information, selection can be worse. We compare these patterns for foreign- and U.S.-educated teams, using only countries where no current manager studied. The key comparison is how stock selection changes with familiarity for each education group.

We measure how closely a market is tied to where a manager studied in three ways, each in a separate regression: a shared border; a common official language; and the log of Facebook social connectedness from the October 2021 snapshot. For foreign-educated managers, the tie is measured between the market and the manager's non-U.S. education countries, using the closest tie when there are several. For U.S.-educated managers, it is measured between the market and the United States. Both groups therefore face the same question: is stock selection better in markets close to where the manager studied?

Ties are defined for managers, but returns are observed for the team, so we average the ties over the team. Let z_{ic} measure how closely market c is tied to where manager i studied: a zero–one indicator for border and language ties, or log social connectedness. For fund j at quarter-end q, with N_{jq} managers of known education, we form two team averages:

$$\text{Tie}_{jcq} = \frac{1}{N_{jq}} \sum_i z_{ic}, \qquad \text{ForeignTie}_{jcq} = \frac{1}{N_{jq}} \sum_i \mathbb{1}_i^{\text{Foreign}} z_{ic},$$

where both sums run over the team's managers with known education and $\mathbb{1}_i^{\text{Foreign}}$ equals one if manager i is foreign-educated. Tie averages every manager's tie; ForeignTie averages the same ties but counts U.S.-educated managers as zero. For example, consider a team with one

[15] Appendix B.8 shows how the foreign-versus-U.S. comparison in these countries changes over the sample period.

manager educated in Germany and one educated only in the United States. For its holdings in Austria, which borders Germany but not the United States, Tie = ForeignTie = 1/2. For its holdings in Canada, which borders the United States but not Germany, Tie = 1/2 and ForeignTie = 0. ForeignTie is therefore an average of manager-level interactions, not the product of the team's foreign share and its average tie. For each tie measure, we regress the fund's adjusted country return in the market, R_{jcq}, on these team averages:

$$\begin{aligned} R_{jcq} = \alpha_{jc} + \lambda_{cq} + \gamma \text{ForeignShare}_{jq} + \delta \text{Tie}_{jcq} \\ + \theta \text{ForeignTie}_{jcq} + \Gamma' X_{jcq} + \varepsilon_{jcq}. \end{aligned} \tag{9}$$

Each coefficient has a direct interpretation. For an entirely U.S.-educated team, only Tie varies with familiarity, so δ describes how stock selection changes with closeness to the United States. For an entirely foreign-educated team, Tie and ForeignTie vary together, so $\delta + \theta$ describes how stock selection changes with closeness to the team's education countries. The focal coefficient θ is the difference between the two slopes; it is negative if closeness is associated with less favorable selection for foreign-educated managers than for U.S.-educated managers. Finally, γ is the foreign-versus-U.S. difference in markets with no tie to either group's education.[16] For the border and language measures, a unit change compares an entirely tied team with an entirely untied team; for social connectedness, slopes are per log point.

Fund-country effects α_{jc} compare the same fund in the same market over time, and country-quarter effects λ_{cq} remove returns common to all funds investing in that market in that quarter. Because each manager's tie to a market is fixed, the slopes are identified from changes in a team's composition within a fund and market, measured relative to other funds investing in the same market that quarter. The sample contains only markets outside the team's current education countries. Controls X contain $\text{MissingShare}_{jcq}$, log country stock

[16] For social connectedness, γ refers to zero log connectedness, outside the observed range, so we interpret only the slopes in that model.

count, log team size, unknown-education share and the five team traits.[17] Countries receive equal weight within each fund-quarter, and standard errors are clustered by fund.

Table 7, column 1, reports the border results. A shared border is worth far less to foreign-educated teams than to U.S.-educated teams: the difference between their slopes, $\theta = -484$ basis points (standard error 158), is significant at 1%, so the same border tie is associated with selection 4.84 percentage points a year less favorable for an entirely foreign-educated team than for an entirely U.S.-educated team. Foreign-educated teams select stocks worse in markets bordering an education country than in other outside markets: their slope $\delta + \theta$ is -184 basis points (standard error 67, significant at 1%).[18] U.S.-educated teams select stocks better in markets bordering the United States: their slope δ is $+300$ basis points (standard error 135, significant at 5%). This U.S.-educated slope rests on only two markets, Canada and Mexico, whereas the foreign-educated slope draws on many country pairs. These are returns per dollar of the country basket, not differences in whole-fund NAV.

The other two measures give the same relative result. With social connectedness (column 3), the difference from U.S.-educated managers is $\theta = -110$ basis points per log point (standard error 42, significant at 1%), or 117 basis points a year for a one-standard-deviation increase in log connectedness; foreign-educated teams' own slope is -43 basis points per log point (standard error 17, significant at 5%). With a common official language (column 2), the difference is $\theta = -231$ basis points (standard error 67, significant at 1%). Here the gap reflects mainly what U.S.-educated teams gain: they select stocks 155 basis points better in markets that share an official language with the United States (standard error 75, significant at 5%), while the foreign-educated slope of -75 basis points is negative but not significant. Each measure enters a separate regression, so the estimates do not depend on conditioning one correlated tie measure on another.

[17] Unavailable pair measures contribute zero to the tie averages, and $\text{MissingShare}_{jcq}$ records the share of managers whose tie cannot be classified: U.S.-educated managers with unclassified geographic ties in the border and language models, and foreign-educated managers with a missing social-connectedness pair in the social model.

[18] Standard errors of $\delta + \theta$ use the joint covariance of the two coefficients from the same regression.

A similar pattern appears when we drop the size and past-return adjustment and compare raw country returns among funds holding the same country in the same quarter (Appendix B.9).

In sum, familiarity is worth much less to foreign-educated teams than to U.S.-educated teams: by all three measures, closeness to where managers studied is associated with less favorable stock selection for foreign-educated teams, by 484 basis points a year for a shared border, 231 for a common official language and 117 for a one-standard-deviation increase in social connectedness. Closeness to an education country thus accompanies poorer stock choices where it would be expected to help, consistent with familiarity bias. Set beside the high-conviction local gains, the result marks the limit of useful local knowledge: it pays in selected education-country stocks but accompanies poorer choices in neighboring and socially connected markets.

4 Recruitment and Career Evaluation

The contrast between local value and broader performance raises a personnel question: do funds distinguish country expertise from investment ability across the mandate when selecting and evaluating managers? Foreign-educated appointments have become more common (Appendix A.1). We first examine whether country losses predict linked hiring and whether local performance subsequently improves. We then compare recruits' prior records. A weaker general investment record could be a rational trade for a stronger fundraising record or a stronger record in the recruit's country specialty, so we examine both. Next, we examine how subsequent fund performance relates to retention and new assignments. A separate flow-and-fee comparison assesses commercial value to the adviser.

4.1 Country performance and the demand for expertise

A fund may seek a manager educated in a country when investing there becomes difficult. We test this possibility using countries already held at the preceding quarter-end.

An observation is a fund j, country c and calendar quarter q, including quarters with no appointment. The outcome indicates that a manager who studied in c joins fund j during q. We express appointment probabilities in basis points: the zero-one indicator is multiplied by 10,000, and one probability basis point equals 0.01 percentage point. We distinguish two prior-year signals: L^M_{cq} is minus the compounded country-market return over the preceding twelve months, and L^P_{jcq} is minus the fund's average country-market-adjusted holdings return over the preceding four quarters. Both are expressed in percentage points, so a larger value means worse performance.[19] We estimate

$$Y^g_{jcq} = \alpha_{jc} + \lambda_{jq} + \beta^g_M L^M_{cq} + \beta^g_P L^P_{jcq} + \Gamma'_g X_{jcq} + \varepsilon^g_{jcq}. \tag{10}$$

Here g denotes the direct schooling link or, in a separate comparison, schooling in a bordering country without schooling in c itself. Fund-country effects remove persistent destination preferences, while fund-quarter effects compare countries within the same fund and calendar quarter. Controls X include the preceding country weight and stock count in logs and the incumbent team's direct and bordering-country schooling shares. Fund-quarter effects also absorb team size and other common fund conditions. We weight countries equally within each fund-quarter and cluster by fund. Country-quarter effects would absorb the country-market signal and are not included.

Both signals predict direct-country hiring (Table 8, column 1). A ten-percentage-point lower prior market return raises the quarterly probability of a linked appointment by 0.43

[19] The market proxy weights the covered stocks by lagged market capitalization, with monthly stock returns winsorized at the 0.1st and 99.9th percentiles. The fund-specific signal averages annualized country-market-adjusted basket returns, requiring at least three of the preceding four payoff quarters. It differs from the size-and-momentum-adjusted country return used in the stock-picking tests. All predictors precede the appointment quarter; no appointment-quarter return enters.

probability basis points, about 3.7% of its average of 11.80 (0.118%); a ten-point worse own country-relative record raises it by 0.48, about 4.1%, holding the market signal constant. Both coefficients are significant at 5%. The comparison uses 978,521 fund-country-quarters at 1,540 funds, with 1,027 linked country appointments. Demand for a schooling link thus rises after disappointing performance in the relevant market. Column 2 asks whether the same losses predict hiring someone educated in a bordering country instead; neither return signal has a clearly estimated association with these appointments, so the response appears specific to a direct schooling link.

After a linked appointment, does country performance improve relative to comparable non-hiring funds, and does improvement extend to the rest of the portfolio? We compare each linked appointment after disappointing performance with up to three funds holding the same country, under the same benchmark and in the same quarter, with similar prior country performance, allocation, stock count and incumbent coverage but no appointment that quarter. We follow both groups for four quarters before and four quarters after appointment, omitting the appointment quarter. Matching precedes the examination of subsequent returns.

To make the local, outside-country and whole-fund comparisons consistent, we use the same observations, controls and fixed effects for all three outcomes. Let a index a matched country-appointment case, j a fund in that case and t the calendar quarter in which returns are earned. We estimate

$$Y_{jat} = \alpha_{ja} + \lambda_{at} + \beta_H \mathbb{1}_{ja}^{\text{Hire}} \times \mathbb{1}_{at}^{\text{Post}} + \Gamma' X_{jt} + \varepsilon_{jat}. \tag{11}$$

Here Y_{jat} is the adjusted return on the focal country's holdings, the adjusted return on holdings outside the hiring team's education countries, or factor-adjusted NAV, each annualized in basis points. The first two outcomes are returns per dollar invested in the respective basket; NAV is a whole-fund return. The education-country set is fixed at appointment and

applied to both hiring and comparison funds throughout the window. Fund-case effects α_{ja} remove persistent differences between the funds, and case-quarter effects λ_{at} remove common changes within the comparison. The indicators identify the hiring fund and the post-appointment quarters. Thus β_H measures the hiring fund's change relative to the change at its comparison funds. Controls X_{jt} are those of the NAV model in Equation (3), including team size, and errors are clustered by economic fund. Appendix B.10 describes matching, weights and support.

The local improvement is economically substantial. On 511 country appointments at 267 hiring funds, adjusted returns in the focal country improve by 228 annualized basket basis points relative to non-hiring funds (standard error 80), significant at 1% (Table 9). This is a 2.28-percentage-point improvement per dollar invested in that country, after the same stock-characteristic adjustment used in the earlier country-return tests. The companion estimates are −15 basis points for outside-country holdings (standard error 34) and +37 for whole-fund NAV (standard error 37). Neither is precisely estimated. The observed improvement is local: the estimates do not establish an offsetting outside loss or an overall NAV gain. Its whole-fund importance depends on the capital invested in that country.

Does this local improvement persist? We extend the comparison through the second year after appointment, retaining the same funds for the pre-appointment year and both subsequent years. In Equation (11), we replace the single hiring-by-post interaction with separate interactions for the first and second years; the pre-appointment year remains the baseline. The fund-case effects, case-quarter effects, controls and fund clustering are unchanged. The two-year requirement yields 430 country appointments at 231 hiring funds, with 514 fund clusters. On this common cohort, the first-year local improvement is 195 annualized basket basis points (standard error 78), whereas the second-year estimate is −11 (standard error 84). The direct second-minus-first-year difference is −205 basis points (standard error 97), significant at 5% (Table 10). The local gain is thus concentrated in the first year and fades in the next. This temporal comparison uses the same appointments in both years; the earlier

228-basis-point estimate uses the larger one-year cohort. The corresponding NAV estimates show no clear change between years. The fading concerns the local payoff, not whole-fund performance.

4.2 Recruits' prior records

The preceding test asks whether the hiring fund improves; we now ask what track record the incoming manager brings. We compare foreign- and U.S.-educated recruits appointed by the same fund, first on their previous teams' investment performance and then on their previous funds' flows. This comparison uses the broader set of experienced appointees, with one observation for each manager's appointment to a fund. We measure the previous teams' average adjusted country return over the four full quarters before hiring, including both education countries and other non-U.S. countries. As in Section 2.4, these returns compare country holdings with similar local stocks. Including education countries allows any demonstrated local advantage to enter the recruit's overall international record; the last test in this subsection examines the education-country record separately.[20]

Let h identify manager i's appointment to fund j in year y. The prior average return, $\bar{R}_h^{\text{pre}}$, is expressed in annualized basis points per dollar of country holdings; the bar denotes an average and pre means before appointment. We first set the outcome Y_h equal to this prior investment record:

$$Y_h = \alpha_j + \lambda_y + \delta \mathbb{1}_i^{\text{Foreign}} + \eta' \mathbf{B}_h + \Gamma' X_h + v_h. \tag{12}$$

The coefficient of interest is δ, on the indicator for a foreign-educated recruit. A negative δ means that foreign-educated recruits had weaker prior team records than U.S.-educated recruits accepted by the same fund, conditional on the controls. Hiring-fund effects α_j

[20] Countries have equal weight within origin fund-quarter. We average quarters, combine share classes of an economic origin and then average distinct origin funds. The characteristic-adjusted record covers non-U.S. stocks and is not a whole-fund NAV return.

account for each fund's persistent recruitment standards, and appointment-year effects λ_y remove common recruitment conditions. The comparison uses hires across years within a fund. Because records partly reflect the previous funds' mandates, $\mathbf{B}_h$ controls for the benchmarks of the recruit's previous funds.[21] Controls X_h include school-ranking categories, gender, CFA, experience, the size and unknown-education share of the origin and destination teams, and the origin teams' country stock count. Appointments receive equal weight, and standard errors are clustered by hiring fund.

The comparison covers experienced appointees whose previous teams have observable records: 1,321 appointments at 300 hiring funds.[22]

Foreign-educated recruits arrive with weaker investment records. Table 11, column 1, estimates a difference of −86 annualized basket basis points (standard error 38), significant at 5%. Their previous teams' adjusted country returns are about 0.86 percentage points lower per year than those of U.S.-educated recruits accepted by the same hiring fund, accounting for origin benchmarks and observed qualifications.

Two finer comparisons in Appendix B.14, which use records outside education countries, give similar gaps. Measuring the previous teams' returns relative to other funds with the same benchmark, country and quarter gives −72 basis points, and comparing only recruits hired by the same fund in the same quarter gives −81; both are significant at 10%.

A weaker investment record could be acceptable if the recruit brings a stronger fundraising record. We test that possibility using flows to the previous funds before appointment. For the prior-record test in column 2, we use $Y_h = \text{Flow}_h^{\text{pre}}$ in Equation (12). This outcome is the quarterly net flow into the recruits' previous funds, in percentage points of beginning-

[21] These are benchmark shares, weighted as in the record, for the 551 appointments with multiple origin benchmarks; Global ex-U.S. is the omitted category among 16. Prior NAV does not enter this equation, so the record is not conditioned on another realized performance measure.

[22] Of the appointments at the 1,663 funds, 1,600 have prior selection records, and support for the paired histories and the fund and year comparisons leaves 1,322 at 300 hiring funds. We use the 1,321 of these that also enter the peer-adjusted outside-country comparison in Appendix B.14, which needs a same-benchmark peer and loses one appointment, so that both records cover the same recruits. The outcome here uses all supported non-U.S. country baskets for these appointments, including education countries, and does not require peer returns. The same-round and education-country record tests use their own samples. Managers without a recorded prior fund role cannot enter this comparison.

of-quarter assets, minus the mean flow of other funds with the same benchmark, averaged over the four quarters before the appointment quarter and across previous funds. Because flows follow returns, this regression controls for the previous funds' benchmark-relative NAV return, $R_h^{\text{NAV,pre}}$, in place of $\mathbf{B}_h$; it uses the 1,322 experienced appointments and otherwise follows Equation (12).[23]

The foreign-education coefficient is -1.8 percentage points per quarter, significant at 10%: relative to U.S.-educated recruits, foreign-educated recruits come from funds with weaker fundraising records even after accounting for prior returns. The weaker investment records thus come with weaker fundraising records, not stronger ones. We next ask whether stronger performance in their education countries compensates for the weaker general investment record.

For this test, we use markets the hiring fund already holds but in which no continuing manager studied, identified from pre-appointment holdings before considering the recruit's education; a recruit educated in such a market adds a schooling link the team lacks.[24] An observation is an appointment h paired with one such country c. The outcome $\bar{R}_{hc}^{\text{pre}}$ is the adjusted return the recruit's previous teams earned on holdings in c over the four full quarters before appointment:[25]

$$\bar{R}_{hc}^{\text{pre}} = \alpha_{jc} + \lambda_{cy} + \beta_G \mathbb{1}_i^{\text{Foreign}} + \beta_L \mathbb{1}_{ic}^{\text{OwnLink}} + \Gamma' X_h + u_{hc}. \tag{13}$$

Here $\mathbb{1}_{ic}^{\text{OwnLink}}$ equals one if the recruit studied in c, so $\beta_G + \beta_L$ compares recruits educated in the country with U.S.-educated recruits appointed by the same fund for the same market. Fund-country and country-year effects and the manager and team controls are included, and

[23] This signal is the mean quarterly relative NAV return in the payoff quarters from five to two quarters before the appointment quarter, in percentage points. It is observed before hiring and does not enter either the main or the peer-adjusted investment-record equation.

[24] Continuing managers are present before and at the end of the appointment quarter. Departures and simultaneous recruits do not supply incumbent coverage. Missing prior holdings remain unclassified.

[25] We average share classes within economic origin-quarter, then quarters and then distinct origins. Every origin fund must have a record in all four quarters. A missing country record remains unavailable rather than becoming zero.

the previous funds' benchmark-relative NAV return replaces the benchmark shares. Each appointment's countries have total weight one, and standard errors are clustered by hiring fund.

Recruits educated in the country bring weaker records there too. The gap $\beta_G + \beta_L$ is -274 annualized basket basis points (standard error 101), significant at 1% (Table 12): about 2.7 percentage points a year on holdings in the very market where the schooling link is relevant.[26] Because it measures records in a single market, on a different sample and with different controls, it does not add to the -86-basis-point general gap. The weaker general record is therefore not offset by a stronger demonstrated record in the potential specialty.

4.3 Careers and performance evaluation

The hiring comparisons show the records funds accept at entry. We now ask how performance relates to managers' careers after appointment: as a fund's returns worsen, are foreign-educated managers less likely to leave relative to U.S.-educated colleagues on the same team? Giving foreign education more weight and realized performance less also predicts a smaller relative gain in new assignments as returns improve. We test both directions among all current managers. This comparison extends the question from recruitment to ongoing evaluation; it is not restricted to the experienced recruits in Section 4.2.

Each observation is a manager i on fund j at quarter-end q. We study two outcomes over the following quarter: $\mathbb{1}_{ijq}^{\text{Exit}}$ equals one if manager i departs from fund j between quarter-ends q and $q+1$, and $\mathbb{1}_{ijq}^{\text{NewFund}}$ equals one if the manager receives a new assignment at another fund. We multiply both indicators by 100, so coefficients are in percentage points. The signal R_{jq}^{rel} is the fund's compounded net return over quarter q minus the mean return of

[26] The sum combines $\beta_G = -55$ and $\beta_L = -219$, using their joint covariance. The comparison uses 9,446 country-appointment records at 1,083 appointments and 257 funds; linked records cover 145 appointments and 85 funds. The 152 linked records span 22 countries, including 72 in the United Kingdom. Starting with 1,131 appointments with supported pre-hire holdings and hiring-fund/year histories, prior-country availability supplies 11,117 records at 1,105 appointments and 269 funds before singleton fixed-effect groups are dropped. Appointments span 2004Q1–2024Q4 and prior payoffs 2003Q1–2024Q3. No fund supplies more than 5.28% of the combined contrast's identifying variation.

other funds sharing its benchmark, also in percentage points. It uses reported whole-fund NAV, covering both U.S. and foreign investments. We estimate one common specification separately for departures and new assignments:

$$Y_{ijq} = \mu_{jq} + \nu_{ij} + \kappa \mathbb{1}_i^{\text{Foreign}} \times R_{jq}^{\text{rel}} + \Gamma' X_{ijq} + e_{ijq}. \tag{14}$$

Here Y_{ijq} is either $100 \times \mathbb{1}_{ijq}^{\text{Exit}}$ or $100 \times \mathbb{1}_{ijq}^{\text{NewFund}}$. The coefficient κ measures how a one-percentage-point improvement in fund returns changes the foreign-minus-U.S. difference in the outcome probability among colleagues on that fund. We denote it by κ_{Exit} for departures and κ_{NewFund} for new assignments. Fund-quarter effects μ_{jq} absorb the common return and fund characteristics shared by colleagues, including the benchmark, mandate and team size. Fund-manager effects ν_{ij} absorb each manager's education and persistent differences in their career prospects at the fund. The coefficient is therefore identified from how colleagues' outcomes diverge as their shared fund return varies.

Controls X_{ijq} include log spell length, log one plus management experience, and interactions of R^{rel} with the female, unknown-gender, CFA and unknown-biography indicators and log experience. Each fund-quarter receives equal total weight, divided equally among its managers, and errors are clustered by economic fund.

Departures indicate whether an existing assignment continues; new assignments indicate additional opportunities. A new assignment is a first appearance on another fund's team in the next quarter, including a new fund launch, and can occur without leaving the original fund. The records track assignment changes without classifying their reasons.

The departure interaction is $\kappa_{\text{Exit}} = +0.145$ percentage points per return percentage point, significant at 1%, on 154,677 manager-quarters at 1,334 funds (Table 13, columns 1–2). A ten-point deterioration lowers the foreign-minus-U.S. departure gap by 1.45 percentage points: foreign-educated managers become less likely to leave relative to their U.S.-educated teammates. For new assignments, $\kappa_{\text{NewFund}} = -0.107$, significant at 5%; a ten-point improve-

ment lowers the foreign-minus-U.S. assignment-probability gap by 1.07 percentage points. The paired signs describe relative protection as returns worsen and a smaller relative gain as returns improve. Both signs are consistent with career evaluations that give the qualification more weight and shared fund returns less. The next comparison asks whether evaluation based on the local specialty can instead explain them.

A specialist may be evaluated on local performance rather than common fund returns. We therefore add the observed performance of education-country holdings and ask whether the foreign-minus-U.S. response to common returns becomes smaller.[27]

Columns 3–4 add this local performance to Equation (14). The local return R^{loc}_{ijq} is the return on the fund's holdings in a foreign-educated manager's education countries over the same quarter, less those countries' index returns, in percentage points. We add it as $\chi \mathbb{1}^{\text{Observed}}_{ijq} \times R^{\text{loc}}_{ijq}$, where $\mathbb{1}^{\text{Observed}}_{ijq}$ equals one for foreign-educated managers whose local return is observed and zero otherwise, including for all U.S.-educated managers, together with the indicator itself, $\phi \mathbb{1}^{\text{Observed}}_{ijq}$. The coefficient χ measures the association between observed local performance and career outcomes, conditional on shared fund performance. If differences in observed local returns account for the groups' different career responses, conditioning on those returns should move κ_{Exit} and κ_{NewFund} toward zero.[28]

Adding local performance leaves the career pattern largely unchanged. The fund-return interactions are +0.138 for departures and −0.108 for assignments, close to the baseline +0.145 and −0.107, and remain statistically significant. The local-return slopes themselves are small and imprecise: 0.017 percentage points for departures and 0.002 for new assignments, per return percentage point. Observed local performance therefore does not account for the different career responses to shared fund returns. Together with the weaker prior education-country records, this result limits an explanation based on demonstrated local

[27] Appendix B.15 examines recorded coverage categories separately.

[28] All 154,677 manager-quarters remain; 48,073 at 832 funds have observed local returns, and the added return term is zero elsewhere. The local return weights holdings by initial country capital. The coefficients χ and ϕ are estimated separately for each outcome, and controls, effects, weights and clustering are unchanged.

expertise at both stages: the record accepted at entry and the performance used in later evaluation.

4.4 Commercial payoff to the adviser

Investment returns are not the only possible source of value to an adviser. Foreign education could attract investors or support higher fees, giving funds a commercial reason to value the background. The prior-flow comparison in Section 4.2 asks what fundraising record recruits bring. Here we ask whether employing more foreign-educated managers is associated with subsequent asset gathering or higher fee rates. The monthly flow regression compares the same fund over time with other funds sharing its benchmark in the same month, accounting for prior performance and fund and team characteristics. The appendix also examines quarterly flows, flows after appointments and reported expense ratios.

In the monthly comparison, a change from an entirely U.S.-educated to an entirely foreign-educated team is associated with monthly net flows 0.21 percentage points higher, with a standard error of 0.15 (Appendix Table A16). Quarterly flows and expense ratios show no clear difference either (Appendix Table A17). These comparisons provide no statistically clear commercial payoff to explain why advisers value the background despite the weaker investment records.

The personnel evidence separates demand for country knowledge from the weight placed on demonstrated performance. Country losses predict linked hiring, and the subsequent local improvement supplies an investment rationale for those appointments. Yet in the broader recruitment sample, foreign-educated managers bring weaker previous-team records both across countries and in their potential specialty. Once managers serve together, their relative career outcomes respond as if less weight were placed on shared fund performance, even after accounting for observed local returns. The commercial comparisons supply no clearly measured compensating payoff. Combined with weaker stock picking in familiar

outside markets, this pattern suggests that funds credit geographic familiarity more broadly than demonstrated investment performance warrants.[29]

5 Conclusion

Geographic familiarity does not translate uniformly into investment value. Within funds over time and across funds sharing a benchmark, greater foreign-educated representation accompanies lower factor-adjusted net and gross returns. Funds overweight education countries without a clear average stock-selection advantage there. High-conviction local stocks earn high alphas under a common country mix, but their estimated alpha contribution at actual capital weights is small. Outside education countries, foreign-educated teams select stocks worse in bordering and more socially connected markets, where familiarity would be expected to help. Their familiarity gradients are also less favorable than those of U.S.-educated teams.

The hiring and career evidence shows how funds respond to this background. Funds seek managers educated in a country after their holdings there underperform. Local stock selection improves relative to comparable non-hiring funds, but the gain fades in the second year; outside-country and overall fund returns show no clear improvement. This gives targeted hiring a local investment rationale. In the broader appointment sample, however, foreign-educated recruits bring weaker prior team investment and fundraising records, including weaker investment records in their potential specialty. Relative to U.S.-educated colleagues, their retention is less sensitive to deteriorating fund performance and their assignment opportunities benefit less as returns improve. Controlling for observed local returns leaves these career differences largely unchanged.

Together, the investment and personnel evidence suggests that funds credit geographic familiarity more broadly than demonstrated investment performance warrants, both in the records accepted at recruitment and in the role of performance in later careers.

[29]The comparisons use recorded benchmarks and accepted recruits' team-level histories. Individual responsibilities, rejected applicants and employers' private information are unobserved.

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Table 1: Summary statistics

	Mean	Median	Std. dev.	N
A. Portfolios and team education (fund-quarter)				
Number of countries held	18.99	19.00	8.44	67,380
Number of manager education countries (including U.S.)	1.57	1.00	0.92	63,510
Number of non-U.S. manager education countries	0.72	0.00	0.95	63,510
B. Funds (fund-year)				
Total net assets ($ billions)	1.59	0.23	6.66	21,084
Net expense ratio (%)	1.22	1.17	0.50	19,695
Turnover ratio (%)	64.31	48.00	60.96	18,999
Annual return (%)	8.68	10.86	22.59	18,773
Annualized volatility (%)	16.46	14.81	7.09	18,773
C. Managers (one observation per fund)				
Average manager tenure (years)	5.57	4.93	3.27	1,663
Number of managers	3.12	2.44	2.44	1,663
D. Foreign education in the net-return panel				
Foreign-educated manager share (fund-month)	0.35			206,847
Fund-quarters with ForeignShare = 0 (%)	48.6			68,949
Fund-quarters with partly unknown education (%)	28.3			68,949
E. Funds with foreign-educated managers (fund-quarter)				
Number of managers	4.13	3.00	4.15	33,761
Number of foreign-educated managers	2.00	1.00	1.72	33,761
Number of foreign countries held	18.44	18.00	8.41	33,761
Foreign countries outside recorded education countries	17.23	17.00	8.06	33,761

Notes: This table reports summary statistics for U.S. actively managed international equity funds (Morningstar), 2000–2024. Panel B summarizes fund characteristics. The NAV regressions use returns and assets of the oldest share class; economic-fund clustering combines identifiers that share a Morningstar fund identifier. Of the 2,992 managers with recorded post-secondary education, 26.9% studied only outside the United States. Panel A counts the distinct education countries of a fund's current managers. Panel E gives equal weight to 33,761 economic fund-quarters at 1,005 funds with positive covered non-U.S. holdings and at least one current manager with a recorded non-U.S. degree. The country and manager counts use the same observations. Outside countries have no schooling link to any current manager in the recorded education histories. ForeignShare is the share of current managers with any recorded non-U.S. education among managers whose education is known.

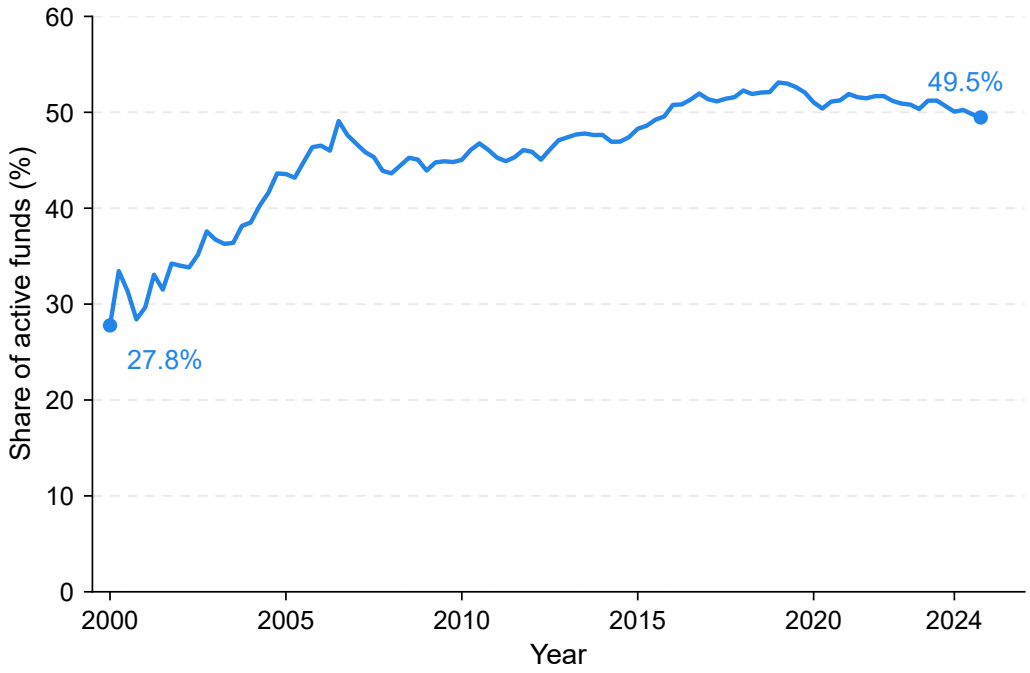


Figure 1: Share of active funds with foreign-educated managers

Notes: Quarterly percentage of active sample funds with at least one current manager with recorded non-U.S. education, 2000Q1–2024Q4. Active funds have positive reported non-U.S. equity holdings in that quarter; no return-availability or stock-classification requirement is imposed. Each economic fund is counted once, with equal weight. Manager status is measured at quarter-end and includes managers educated both in the United States and abroad. All active funds remain in the denominator; unknown education is not classified as U.S.-only. Changes in education coverage can affect the trend. The endpoints are 85 of 306 funds (27.8%) and 369 of 746 funds (49.5%).

Table 2: Foreign education and net fund returns

Dependent variable	Factor-adjusted net NAV return			
Units	Annualized bps			
	(1)	(2)	(3)	(4)
Specification	Full controls	Broader sample	No operating controls	Full controls (WLS)
Foreign-educated share	-69.51^{**}	-69.30^{*}	-68.31^{*}	-69.66^{***}
	(35.42)	(35.43)	(35.61)	(22.66)
Log team size	−88.89	−87.85	−90.36	-72.93^{*}
	(59.10)	(59.06)	(59.22)	(37.80)
Unknown-education share	−108.22	−108.32	−100.94	-134.06^{***}
	(68.59)	(68.54)	(68.22)	(44.16)
Log lagged assets (USD millions)	-21.83^{**}	-22.38^{**}	-21.14^{**}	-17.30^{**}
	(9.90)	(9.84)	(9.91)	(7.38)
Assets spline basis 1	-100.49^{***}	-99.74^{***}	-99.81^{***}	-62.79^{***}
	(26.04)	(25.98)	(26.16)	(18.26)
Assets spline basis 2	559.63^{***}	556.76^{***}	553.00^{***}	327.21^{***}
	(158.52)	(158.37)	(158.96)	(106.46)
Team-size spline basis 1	−78.04	−79.95	−77.85	−24.61
	(137.48)	(137.40)	(137.52)	(88.00)
Team-size spline basis 2	695.54	701.71	686.45	394.77
	(590.07)	(589.75)	(589.82)	(374.04)
Female share (known gender)	−50.70	−50.08	−50.77	−37.88
	(38.54)	(38.52)	(38.58)	(27.60)
Unknown-gender share	−89.18	−90.07	−93.77	−0.34
	(93.07)	(93.05)	(92.41)	(53.01)
CFA share (observed biographies)	6.58	6.77	7.99	33.02
	(31.47)	(31.47)	(31.48)	(21.24)
Unknown-biography share	227.18^{**}	228.44^{**}	227.79^{**}	80.74
	(107.77)	(107.71)	(107.52)	(66.83)
Mean log(1 + experience)	-34.70^{**}	-34.15^{**}	-35.16^{**}	-39.13^{***}
	(16.75)	(16.73)	(16.88)	(10.63)
Prior-year relative NAV return (pp)	-4.60^{***}	-4.77^{***}	-4.63^{***}	-2.92^{***}
	(1.46)	(1.46)	(1.47)	(0.95)
Log(1 + prior relative volatility)	62.22^{**}	63.78^{**}	63.77^{**}	36.75^{**}
	(31.00)	(30.88)	(31.67)	(18.59)
Prior-quarter flow (pp)	1.57×10^{-6}	1.54×10^{-6}	1.55×10^{-6}	1.21×10^{-6}
	(2.25×10^{-6})	(2.24×10^{-6})	(2.24×10^{-6})	(1.51×10^{-6})
Missing prior return history	99.32	103.71^{*}	75.70	2.29
	(62.86)	(62.40)	(57.18)	(38.49)
Missing prior-quarter flow	−41.44	−38.72	−43.26	-66.19^{*}
	(55.66)	(53.41)	(55.30)	(36.44)
Lagged expense ratio (bps)	−0.40	−0.39	—	-0.46^{**}
	(0.34)	(0.34)		(0.23)
Lagged log turnover	84.01^{**}	82.51^{**}	—	30.18
	(35.50)	(35.41)		(25.07)
Top-ten equity concentration	75.23	68.69	—	79.59
	(138.23)	(137.84)		(90.84)
Missing lagged expense ratio	-111.04^{**}	-111.35^{**}	—	−23.98
	(54.10)	(53.80)		(34.88)
Missing lagged turnover	72.29^{**}	71.47^{*}	—	10.76
	(36.75)	(36.70)		(23.92)
Missing equity concentration	26.85	25.74	—	18.74
	(42.14)	(42.13)		(27.61)
Fund FE	Yes	Yes	Yes	Yes
Benchmark × month FE	Yes	Yes	Yes	Yes
Observations	206,847	208,425	206,847	206,847
Fund clusters	1,418	1,536	1,418	1,418

Notes: Monthly net NAV uses thirteen predetermined factor exposures, January 2000–December 2024. Coefficients are annualized bps; foreign share compares one and zero among known-schooling managers. Column 2 adds shorter-history funds with benchmark-peer loadings. Column 3 omits expenses, turnover and concentration. Column 4 retains column 1's sample and controls with inverse fund residual-variance weights; columns 1–3 give fund-months equal weight. Prior flow is uncapped and in raw percentage points, with a missingness indicator. Outcomes are not winsorized or zero-filled. Fund-clustered standard errors appear in parentheses below coefficients. *, ** and *** denote two-sided significance at 10%, 5% and 1%.

Table 3: Fees and the net-return shortfall

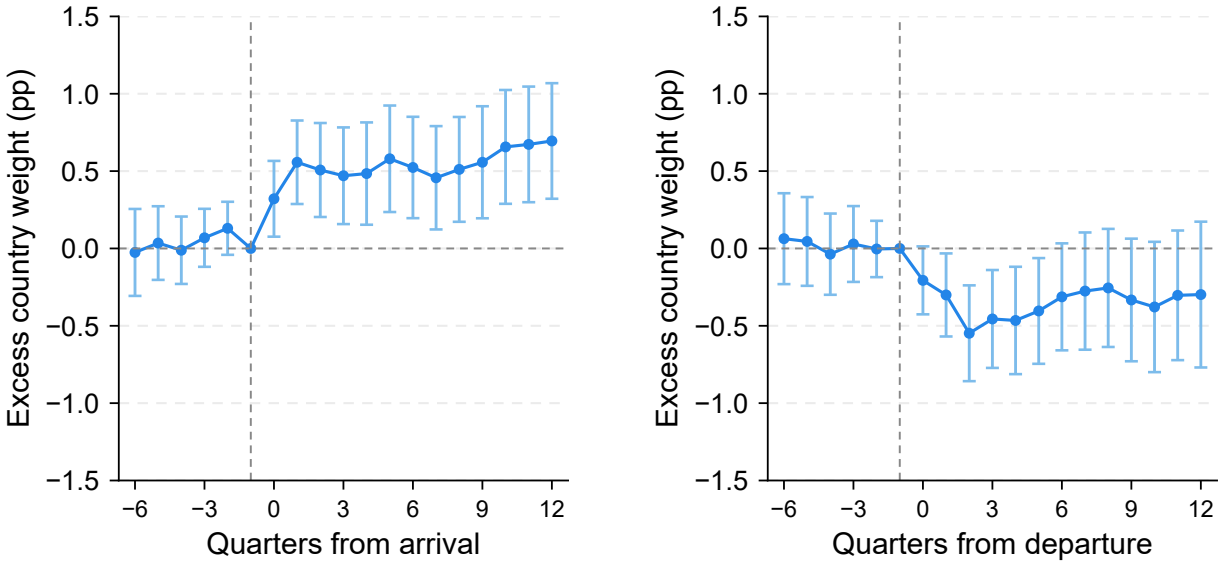
Dependent variable	Net NAV	Gross NAV	Net NAV	Gross NAV
Units		Factor-adjusted returns, annualized bps		
	(1)	(2)	(3)	(4)
Specification	Full controls	Full controls	No operating controls	No operating controls
Foreign-educated share	-70.56^{**}	-71.03^{**}	-69.29^{*}	-70.13^{**}
	(35.42)	(35.40)	(35.62)	(35.57)
Log team size	-98.60^{*}	-102.57^{*}	-99.66^{*}	-104.30^{*}
	(59.39)	(59.30)	(59.51)	(59.39)
Unknown-education share	−110.61	−112.41	−102.99	−105.70
	(68.97)	(69.14)	(68.57)	(68.76)
Log lagged assets (USD millions)	-21.41^{**}	-21.92^{**}	-20.73^{**}	-21.13^{**}
	(10.01)	(10.03)	(10.03)	(10.04)
Assets spline basis 1	-99.72^{***}	-102.85^{***}	-99.14^{***}	-103.73^{***}
	(26.23)	(26.23)	(26.36)	(26.34)
Assets spline basis 2	551.43^{***}	567.58^{***}	544.25^{***}	564.86^{***}
	(159.79)	(159.72)	(160.21)	(160.07)
Team-size spline basis 1	−53.74	−48.77	−55.10	−49.90
	(137.51)	(137.24)	(137.46)	(137.12)
Team-size spline basis 2	598.98	590.68	597.53	592.21
	(590.13)	(589.18)	(589.31)	(588.13)
Female share (known gender)	−54.16	−54.52	−53.78	−54.23
	(38.51)	(38.51)	(38.57)	(38.56)
Unknown-gender share	−89.22	−89.81	−94.35	−95.36
	(93.32)	(93.38)	(92.67)	(92.86)
CFA share (observed biographies)	6.40	5.06	8.11	6.54
	(31.83)	(31.88)	(31.86)	(31.91)
Unknown-biography share	243.98^{**}	248.18^{**}	245.10^{**}	250.14^{**}
	(108.10)	(108.14)	(107.86)	(107.95)
Mean log(1 + experience)	-34.30^{**}	-35.04^{**}	-34.56^{**}	-35.64^{**}
	(16.81)	(16.80)	(16.93)	(16.92)
Prior-year relative NAV return (pp)	-4.46^{***}	-4.50^{***}	-4.48^{***}	-4.50^{***}
	(1.48)	(1.48)	(1.48)	(1.48)
Log(1 + prior relative volatility)	54.63^{*}	55.14^{*}	56.26^{*}	57.68^{*}
	(31.13)	(31.17)	(31.82)	(31.85)
Prior-quarter flow (pp)	1.43×10^{-6}	1.47×10^{-6}	1.40×10^{-6}	1.47×10^{-6}
	(2.25×10^{-6})	(2.26×10^{-6})	(2.24×10^{-6})	(2.25×10^{-6})
Missing prior return history	107.10^{*}	104.68	78.87	78.88
	(63.81)	(63.90)	(57.63)	(57.65)
Missing prior-quarter flow	−67.65	−68.84	−69.12	−70.44
	(55.95)	(56.00)	(55.76)	(55.81)
Lagged expense ratio (bps)	−0.36	−0.07	—	—
	(0.34)	(0.35)		
Lagged log turnover	83.38^{**}	82.62^{**}	—	—
	(35.68)	(35.65)		
Top-ten equity concentration	71.59	67.49	—	—
	(139.26)	(139.32)		
Missing lagged expense ratio	-120.73^{**}	−84.63	—	—
	(53.46)	(54.45)		
Missing lagged turnover	80.45^{**}	80.34^{**}	—	—
	(35.65)	(35.75)		
Missing equity concentration	32.49	30.96	—	—
	(42.37)	(42.49)		
Fund FE	Yes	Yes	Yes	Yes
Benchmark × month FE	Yes	Yes	Yes	Yes
Observations	205,773	205,773	205,773	205,773
Fund clusters	1,418	1,418	1,418	1,418

Notes: Net and reported gross NAV use identical observations and thirteen predetermined factor exposures, 2000–2024. Columns 1–2 use full controls; 3–4 omit expense, turnover and concentration controls. Equal monthly weights; raw prior flow with a missing indicator; economic-fund clustering. Standard errors appear in parentheses below coefficients. *, ** and *** denote two-sided significance at 10%, 5% and 1%. Gross-minus-net foreign-share differences are −0.47 (SE 1.01) for columns 2 minus 1 and −0.84 (SE 1.33) for columns 4 minus 3; paired fund-influence covariance is used.

Table 4: Education links and excess country weights

Dependent variable	Country overweight			
Units	Percentage points			
	(1)	(2)	(3)	(4)
Specification	Any link	Link share	Any link	Link share
Any manager educated in country	1.108***	—	1.091***	—
	(0.220)		(0.219)	
Share of managers educated in country	—	2.626***	—	2.590***
		(0.459)		(0.454)
Female share (known gender)	0.019	0.025	—	—
	(0.043)	(0.042)		
Unknown-gender share	0.087	0.085	—	—
	(0.060)	(0.059)		
CFA share (observed biographies)	−0.030	−0.032	—	—
	(0.032)	(0.032)		
Unknown-biography share	−0.116	−0.122	—	—
	(0.080)	(0.080)		
Mean log(1 + experience)	0.014	0.013	—	—
	(0.014)	(0.014)		
Missing team-trait record	−0.060	−0.067	—	—
	(0.103)	(0.103)		
Fund FE	Yes	Yes	No	No
Country × month FE	Yes	Yes	Yes	Yes
Fund × month FE	No	No	Yes	Yes
Observations	5,613,030	5,613,030	5,611,449	5,611,449
Fund clusters	1,663	1,663	1,661	1,661

Notes: The outcome is country weight minus benchmark weight, in pp, January 1999–December 2024. Any link is an indicator; link share is the fraction of managers educated in the country. Fund-month effects in columns 3–4 absorb team traits, shown as dashes rather than zeros. Observations have equal weight. Standard errors, clustered by fund identifier, appear in parentheses below coefficients. *, ** and *** denote two-sided significance at 10%, 5% and 1%.



(a) Arrival of the first education-linked manager

(b) Departure of the last education-linked manager

Figure 2: Excess country weight around manager arrivals and departures

Notes: This figure plots event-study coefficients for the fund's country weight minus the benchmark's country weight (percentage points) around the quarter in which the fund gains its first manager with non-U.S. education in the country (left) or loses the last such manager (right). Windows run up to twelve quarters and end if the status reverses; events without a six-quarter clean pre-period and overlapping windows are removed (977 arrivals and 793 departures). Regressions include fund × country, country × quarter and fund × quarter fixed effects, with quarter −1 as the reference. Bars are 95% confidence intervals with standard errors clustered by fund.

Table 5: Stock selection in education countries relative to U.S.-educated teams

Dependent variable	Adjusted country return	
Units	Annualized bps	
	(1)	(2)
Specification	OLS	WLS
Foreign-educated share (γ, outside gap)	−46.31	−31.20
	(31.35)	(28.87)
Education-country link (β, local increment)	46.52	36.77
	(49.90)	(47.44)
Log country stock count	−57.23***	−52.91***
	(11.42)	(11.36)
Log team size	−67.97***	−38.56*
	(21.46)	(19.84)
Unknown-education share	25.76	2.85
	(84.74)	(78.94)
Female share (known gender)	−96.56**	−117.46***
	(44.95)	(37.52)
Unknown-gender share	94.11	132.75
	(86.85)	(81.54)
CFA share (observed biographies)	51.48*	47.34
	(30.20)	(29.64)
Unknown-biography share	−39.73	−45.65
	(123.43)	(115.07)
Mean log(1 + experience)	−52.48***	−43.23***
	(15.43)	(15.14)
Fund × country FE	Yes	Yes
Country × quarter FE	Yes	Yes
Observations	974,924	974,924
Fund clusters	1,428	1,428

Notes: Equation (6) uses all held foreign countries and the adjusted country return of Equation (2) over the next three months. $\mathbb{1}^{\text{Link}} = 1$ if a current manager studied in the country; ForeignShare is the fraction of known-education managers with foreign schooling. The linked-country comparison sets $\mathbb{1}^{\text{Link}} = 1, \text{ForeignShare} = 1$ against $\mathbb{1}^{\text{Link}} = 0, \text{ForeignShare} = 0$. The $\mathbb{1}^{\text{Link}}$ slope alone is not that comparison. Countries have equal weight within each fund-quarter; WLS also weights by inverse fund residual variance. Outcomes cover January 2000–December 2024; 1,429 identifiers map to 1,428 economic-fund clusters. Standard errors appear in parentheses below coefficients. *, ** and *** denote two-sided significance at 10%, 5% and 1%. The focal comparison is $\beta+\gamma$: an entirely foreign-educated team investing in an education country versus an entirely U.S.-educated team in the same market. The estimates are 0.21 (SE 56.61) and 5.57 (51.39) bps in columns 1 and 2, using joint covariance.

Table 6: Local expertise and high-conviction stock selection

Dependent variable	Portfolio excess return			
Units	Annual alpha in bps; factor loadings dimensionless			
	(1)	(2)	(3)	(4)
Specification	Local top	Local ordinary	Outside top	Outside ordinary
Alpha (annualized bps)	400.2^{***}	40.1	13.4	38.1
	(133.3)	(95.9)	(54.8)	(39.9)
Global ex-U.S. benchmark excess return	0.148	0.135^{*}	0.208^{***}	0.183^{***}
	(0.106)	(0.076)	(0.043)	(0.032)
Developed ex-U.S. market	0.778^{***}	0.723^{***}	0.638^{***}	0.647^{***}
	(0.097)	(0.069)	(0.040)	(0.029)
Developed ex-U.S. size	−0.012	-0.146^{***}	0.059^{**}	-0.048^{***}
	(0.059)	(0.043)	(0.024)	(0.018)
Developed ex-U.S. value	−0.011	0.021	0.063^{**}	−0.013
	(0.068)	(0.049)	(0.028)	(0.020)
Developed ex-U.S. investment	0.030	−0.059	-0.080^{**}	−0.029
	(0.093)	(0.067)	(0.038)	(0.028)
Developed ex-U.S. profitability	0.015	−0.057	0.055	−0.008
	(0.093)	(0.067)	(0.038)	(0.028)
Emerging-market market	0.103^{**}	0.175^{***}	0.192^{***}	0.183^{***}
	(0.044)	(0.031)	(0.018)	(0.013)
Emerging-market size	0.000	−0.029	0.023	-0.040^{**}
	(0.064)	(0.046)	(0.026)	(0.019)
Emerging-market value	−0.011	−0.019	-0.065^{**}	-0.049^{**}
	(0.066)	(0.047)	(0.027)	(0.020)
Emerging-market investment	−0.041	0.001	0.129^{***}	−0.020
	(0.077)	(0.055)	(0.032)	(0.023)
Emerging-market profitability	−0.064	0.041	0.073^{**}	0.072^{***}
	(0.085)	(0.061)	(0.035)	(0.025)
Observations	300	300	300	300

Notes: Quarter-end active stock weights identify top-quartile and ordinary holdings. Portfolios are value-weighted and reweighted to benchmark country shares, January 2000–December 2024. Annual alpha is $12 \times 10^4 \alpha_p$ basis points; factor loadings are dimensionless. Individual regressions use conventional OLS standard errors. Direct alpha differences and Wald tests use the robust stacked-system covariance without calendar-month clustering. Benchmark annual excess return is 4.30%. Standard errors appear in parentheses below coefficients. *, ** and *** denote two-sided significance at 10%, 5% and 1%. The local top-minus-ordinary alpha is 360.1 bps (SE 137.1, $p = 0.009$); the outside counterpart is −24.7 (SE 60.3, $p = 0.682$). Local-minus-outside equality tests give $p = 0.002$ for top and $p = 0.981$ for ordinary portfolios. Figure 3 displays these four tests.

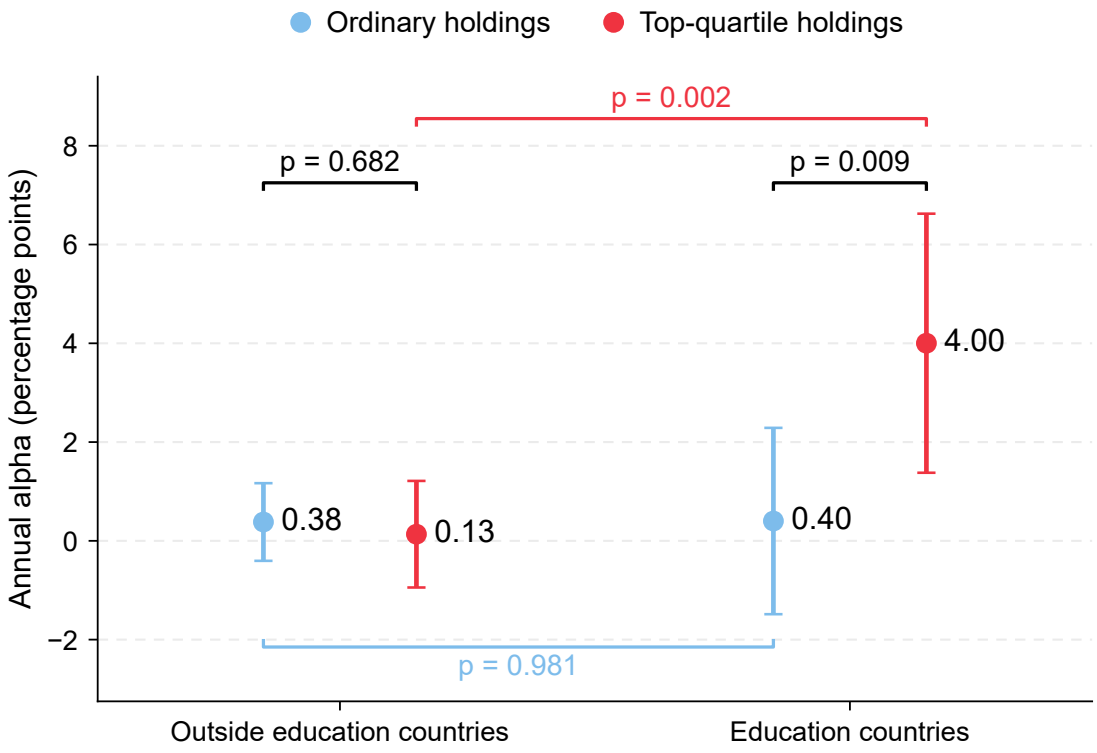
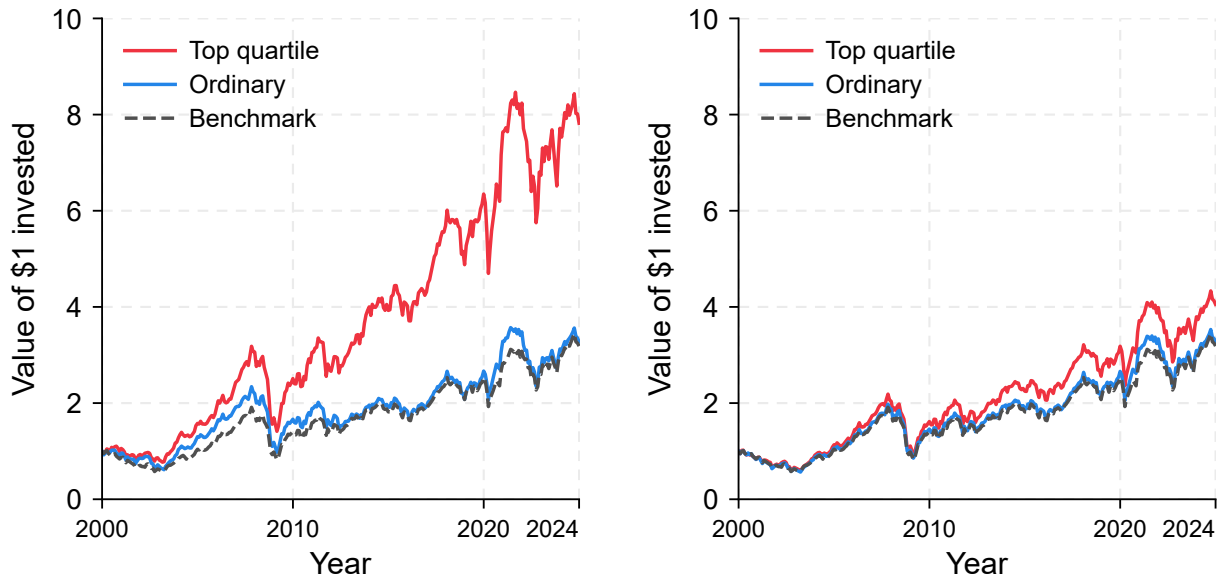


Figure 3: Conviction sorting: portfolio alphas and direct comparisons

Notes: The figure displays the four regressions in Table 6, using January 2000–December 2024. Alpha and its 95% confidence interval are annualized percentage points. Top denotes the highest within-country I-measure quartile; ordinary denotes the other three quartiles. The brackets report all four Wald tests: top versus ordinary within each country group, and education versus outside countries within each conviction group. Individual intervals use conventional OLS inference; differences use a robust stacked covariance. Country portfolios are value-weighted and reweighted to the common benchmark country mix.



(a) Education countries

(b) Outside education countries

Figure 4: Cumulative returns of conviction-sorted portfolios

Notes: Each panel compounds the monthly gross returns of the same top and ordinary strategies used in Table 6. One dollar is invested immediately before January 2000; all 300 monthly returns through December 2024 are included. The dashed line is the common Morningstar Global ex-U.S. benchmark. These are returns on country-reweighted disclosed-holdings strategies, not actual fund NAV, capital contributions or cumulative risk-adjusted alpha. The two panels use the same vertical scale.

Table 7: Relatedness and stock selection outside education countries

Dependent variable	Adjusted country return		
Units	Annualized bps		
	(1)	(2)	(3)
Specification	Shared border	Official language	Social connectedness
Foreign education × tie (θ)	-484.20^{***}	-230.69^{***}	-109.98^{***}
	(158.36)	(67.31)	(42.05)
Tie (δ)	299.90^{**}	155.35^{**}	67.17
	(134.65)	(75.45)	(43.53)
Foreign-educated share (γ)	−28.66	1.14	826.14^{**}
	(32.33)	(33.18)	(330.63)
Missing-tie share (MissingShare)	8.29	38.62	-333.29^{*}
	(45.48)	(45.98)	(174.15)
Log country stock count	-62.09^{***}	-62.27^{***}	-65.81^{***}
	(11.73)	(11.73)	(11.60)
Log team size	-66.44^{***}	-67.15^{***}	-60.96^{***}
	(21.32)	(21.32)	(22.16)
Unknown-education share	59.19	60.97	43.81
	(84.35)	(85.02)	(83.59)
Female share (known gender)	-95.46^{**}	-96.27^{**}	-103.92^{**}
	(45.61)	(45.61)	(47.46)
Unknown-gender share	54.04	47.36	61.07
	(69.76)	(69.79)	(71.85)
CFA share (observed biographies)	43.11	42.28	44.36
	(30.85)	(30.91)	(31.05)
Unknown-biography share	22.70	30.46	7.04
	(119.49)	(119.55)	(114.17)
Mean log(1 + experience)	-44.05^{***}	-44.47^{***}	-38.68^{**}
	(15.57)	(15.60)	(16.04)
Fund × country FE	Yes	Yes	Yes
Country × quarter FE	Yes	Yes	Yes
Observations	936,076	936,076	876,532
Fund clusters	1,423	1,423	1,421

Notes: Equation (9); the tie averages are defined in Section 3.5. Each column uses one familiarity measure. Column 1 is a shared border; column 2 is a common official language; column 3 is log social connectedness from the October 2021 snapshot, without standardization or centering. Its education-share and missing-share coefficients refer to log connectedness zero; at mean log connectedness 8.107, the comparable education-share level is −65.38 bps. The text translates the raw gradient into a standard-deviation comparison. Foreign-educated managers' ties use their non-U.S. schooling countries; U.S.-educated managers' ties use the United States. Rows report the coefficients of actual regressors in Equation (9). The first row is θ, the coefficient on ForeignTie: the team average of each manager's foreign-education indicator interacted with that manager's tie, not foreign share times the team-average tie. It directly tests the foreign-minus-U.S. difference in slopes. The tie coefficient δ gives the U.S.-educated slope; the foreign-educated slope is their sum. All direct schooling destinations are excluded. Countries receive equal weight within fund-quarter. Standard errors are clustered by economic fund. Samples contain 1,424/1,422 ISINs and 1,423/1,421 economic fund clusters for discrete/SCI measures. Standard errors appear in parentheses below coefficients. *, ** and *** denote two-sided significance at 10%, 5% and 1%.

Table 8: Country performance and education-related hiring

Dependent variable	Direct-country appointment	Bordering-country appointment
Units	Quarterly appointment probability, bps	
	(1)	(2)
Prior country-market loss (pp)	0.043**	−0.014
	(0.021)	(0.028)
Prior own country underperformance (pp)	0.048**	0.036
	(0.024)	(0.023)
Log prior country weight	1.226	−1.227
	(0.818)	(0.920)
Log prior country stock count	0.834	2.061
	(1.520)	(1.527)
Incumbent direct-schooling share	−344.225***	5.472
	(40.964)	(14.110)
Incumbent foreign-border-schooling share	5.629	−399.461***
	(13.202)	(43.656)
Fund × country FE	Yes	Yes
Fund × quarter FE	Yes	Yes
Observations	978,521	978,521
Fund clusters	1,540	1,540

Notes: Equation (10). Appointments use recorded schooling at all degree levels and dated spell starts. Column 1 indicates an entrant educated in the held country. Column 2 indicates a foreign-educated entrant who studied in a bordering country but not in the held country. Both outcomes are indicators multiplied by 10,000, so coefficients are probability basis points; one basis point is 0.01 percentage point. Return predictors are in percentage points. The loss measures are minus prior returns in percentage points; positive values mean weaker performance. Country-market returns compound twelve preceding months; own underperformance averages annualized country-market-adjusted returns over the preceding four quarters, requiring at least three. The remaining controls are measured before appointment. Incumbent shares divide by all team members; team size and overall missing education are absorbed by fund-quarter effects. Each fund-quarter receives equal weight across its countries. Appointments span 2001Q1–2024Q4. Standard errors appear in parentheses below coefficients and are clustered by economic fund. *, ** and *** denote two-sided significance at 10%, 5% and 1%.

Table 9: Local, outside-country and NAV performance following linked appointments

Dependent variable	Adjusted local return	Adjusted outside return	Factor-adjusted NAV
Units	Basket bps/year	Basket bps/year	Fund bps/year
	(1)	(2)	(3)
Linked appointment × Post	228.18***	−14.94	36.82
	(80.48)	(33.93)	(37.37)
Log team size	−945.39**	−583.91***	−200.67
	(460.95)	(176.73)	(242.97)
Unknown-education share	1427.32*	6.06	496.76**
	(763.87)	(225.07)	(247.31)
Log lagged assets (USD millions)	−20.11	−113.83	−114.38*
	(115.84)	(70.79)	(69.29)
Assets spline basis 1	−103.60	60.01	−96.08
	(288.40)	(136.72)	(167.19)
Assets spline basis 2	−591.91	−896.10	−33.99
	(1823.30)	(756.91)	(937.12)
Team-size spline basis 1	928.86	938.84**	403.82
	(1005.21)	(383.95)	(523.96)
Team-size spline basis 2	−2699.45	−3006.56*	−1414.81
	(4093.10)	(1584.55)	(2156.91)
Female share (known gender)	−69.39	−16.20	8.81
	(320.13)	(146.05)	(165.84)
Unknown-gender share	−1155.27*	−368.35	−377.52
	(614.03)	(318.51)	(245.47)
CFA share (observed biographies)	346.36	160.48*	190.81
	(270.98)	(96.26)	(135.54)
Unknown-biography share	250.16	264.22	−49.08
	(894.74)	(371.03)	(325.80)
Mean log(1 + experience)	−290.48**	4.85	−56.31
	(142.55)	(50.04)	(64.65)
Prior-year relative NAV return (pp)	−25.32***	−27.24***	−35.58***
	(8.25)	(3.41)	(3.86)
Log(1 + prior relative volatility)	307.59	−55.50	−81.62
	(210.11)	(72.74)	(82.49)
Prior-quarter flow (pp)	-8.79×10^{-5}	$-3.56 \times 10^{-4***}$	$-2.00 \times 10^{-4***}$
	(1.84×10^{-4})	(7.67×10^{-5})	(6.24×10^{-5})
Missing prior return history	302.45	16.42	131.43
	(501.24)	(181.38)	(205.46)
Missing prior-quarter flow	169.39	−225.40	−451.38*
	(504.41)	(243.61)	(242.37)
Lagged expense ratio (bps)	3.17	−1.36	−0.19
	(3.93)	(1.13)	(1.65)
Lagged log turnover	94.77	161.03	358.33***
	(251.56)	(103.32)	(108.65)
Top-ten equity concentration	−904.39	−371.90	290.82
	(1160.77)	(553.58)	(565.07)
Missing lagged expense ratio	307.28	−121.37	−322.34
	(487.00)	(193.21)	(250.14)
Missing lagged turnover	15.94	33.22	145.02
	(283.24)	(112.15)	(119.25)
Fund × matched-case FE	Yes	Yes	Yes
Matched-case × quarter FE	Yes	Yes	Yes
Observations	13,237	13,237	13,237
Country-appointment cases	511	511	511
Hiring funds	267	267	267
Fund clusters	610	610	610

Notes: Equation (11). All columns use identical observations and controls. Linked appointments follow disappointing country-market or own country-relative performance. Comparison funds hold the same country under the same benchmark in the same appointment quarter, matched on prior conditions, with no appointment that quarter. Pre- and post-periods each cover four quarters, excluding quarter zero and requiring at least three common quarters per side. Geography is fixed at appointment for both groups. Equal hiring-fund-quarter weights are divided across cases, hiring/comparison sides, phases and observed quarters. The concentration-missingness control is absorbed. Basket returns are characteristic-adjusted; NAV uses the factor adjustment of Table 2. Basket coefficients have separate capital denominators and do not add to NAV. Returns end in 2024Q4. Fund-clustered standard errors appear in parentheses below coefficients. *, ** and *** denote two-sided significance at 10%, 5% and 1%.

Table 10: Persistence of the local gain following linked appointments

Dependent variable Units	Adjusted local return Basket bps/year (1)	Factor-adjusted NAV Fund bps/year (2)
Linked appointment × Year 1	194.80**	−13.90
	(78.39)	(38.08)
Linked appointment × Year 2	−10.54	12.12
	(83.83)	(40.61)
Linear combination of the two appointment coefficients		
Year 2 minus Year 1	−205.34**	26.03
	(96.72)	(34.02)
Log team size	−551.22	−91.36
	(347.78)	(225.61)
Unknown-education share	1274.38**	390.96
	(499.16)	(237.49)
Log lagged assets (USD millions)	−56.86	−40.56
	(99.45)	(52.69)
Assets spline basis 1	99.46	−218.64*
	(231.40)	(124.96)
Assets spline basis 2	−1878.87	757.42
	(1395.73)	(725.05)
Team-size spline basis 1	281.64	−141.41
	(879.99)	(435.46)
Team-size spline basis 2	−889.26	921.12
	(3750.55)	(1707.10)
Female share (known gender)	−110.56	−26.70
	(271.66)	(134.32)
Unknown-gender share	−517.29	−195.35
	(486.74)	(179.67)
CFA share (observed biographies)	329.78	105.77
	(217.80)	(97.87)
Unknown-biography share	−153.07	−255.32
	(624.31)	(279.48)
Mean log(1 + experience)	−302.30***	−118.01**
	(110.66)	(55.58)
Prior-year relative NAV return (pp)	−22.98***	−29.75***
	(7.56)	(3.66)
Log(1 + prior relative volatility)	173.83	−23.87
	(152.43)	(67.05)
Prior-quarter flow (pp)	-1.12×10^{-4}	-1.40×10^{-4}**
	(1.68×10^{-4})	(5.45×10^{-5})
Missing prior return history	79.59	136.21
	(518.35)	(189.33)
Missing prior-quarter flow	−284.37	−887.02***
	(597.20)	(292.45)
Lagged expense ratio (bps)	0.91	0.11
	(3.07)	(1.45)
Lagged log turnover	−54.73	304.88***
	(213.12)	(109.01)
Top-ten equity concentration	−1821.10*	171.01
	(1013.13)	(487.20)
Missing lagged expense ratio	479.59	−219.53
	(449.77)	(225.04)
Missing lagged turnover	−165.75	89.68
	(198.16)	(127.18)
Fund × matched-case FE	Yes	Yes
Matched-case × quarter FE	Yes	Yes
Observations	15,862	15,862
Country-appointment cases	430	430
Hiring funds	231	231
Fund clusters	514	514

Notes: Equation (11) with separate hiring interactions for quarters +1 to +4 and +5 to +8; quarters −4 to −1 are the baseline and quarter zero is omitted. Both columns use the same cases, observations and controls, with the matched non-hiring funds of Table 9, and require at least three observed quarters in each phase. Appointments end by 2022Q4 and returns by 2024Q4. Equal hiring-fund-quarter weights are divided across cases, hiring/comparison sides, phases and observed quarters. The concentration-missingness control is absorbed. The year difference uses the joint covariance. Standard errors appear in parentheses below coefficients and are clustered by fund. *, ** and *** denote two-sided significance at 10%, 5% and 1%.

Table 11: Prior records at appointment

Dependent variable	Prior adjusted return	Prior fund flow
Units	Annualized basket bps	Quarterly pp
	(1)	(2)
Foreign-educated appointee	−86.41**	−1.83*
	(38.18)	(1.06)
Log pre-hire destination team size	6.17	0.30
	(65.17)	(1.02)
Pre-hire destination unknown education	246.69	1.10
	(172.46)	(2.91)
Log origin team size	−13.50	−2.83**
	(49.99)	(1.18)
Origin-team unknown education	66.20	−7.84
	(287.65)	(7.12)
Log origin country stock count	−80.15	−1.47
	(63.12)	(1.14)
Appointee: top-100 school	54.42	−0.89
	(35.41)	(0.73)
Appointee: school ranked 101–500	106.35**	0.10
	(50.63)	(1.10)
Appointee: female	−21.00	0.83
	(33.75)	(0.85)
Appointee: unknown gender	58.50	1.30
	(85.40)	(2.53)
Appointee: CFA	−56.18*	1.91**
	(31.32)	(0.78)
Log(1 + manager experience)	50.49*	−1.89***
	(29.18)	(0.60)
Prior origin relative NAV (pp)	—	1.73***
		(0.47)
Prior benchmark: China	238.93	—
	(246.40)	
Prior benchmark: Developed Asia-Pacific	−1851.57*	—
	(969.62)	
Prior benchmark: Developed Asia-Pacific ex Japan	238.91*	—
	(139.68)	
Prior benchmark: Developed Europe	569.03***	—
	(193.48)	
Prior benchmark: Emerging Americas	−358.95**	—
	(176.43)	
Prior benchmark: Emerging markets	34.22	—
	(104.25)	
Prior benchmark: Global growth	336.90*	—
	(192.76)	
Prior benchmark: Global large cap	−55.15	—
	(242.52)	
Prior benchmark: Global	−82.44	—
	(127.64)	
Prior benchmark: Global small/mid cap	14.81	—
	(187.87)	
Prior benchmark: Global value	−107.66	—
	(228.04)	
Prior benchmark: Global ex-U.S. growth	7.68	—
	(120.20)	
Prior benchmark: Global ex-U.S. small/mid cap	180.53	—
	(135.85)	
Prior benchmark: Global ex-U.S. value	−170.82	—
	(112.42)	
Prior benchmark: Japan	3214.77	—
	(2386.77)	
Hiring-fund FE	Yes	Yes
Appointment-year FE	Yes	Yes
Observations	1,321	1,322
Hiring-fund clusters	300	300

Notes: Equation (12). Reference group: U.S.-educated appointees. Returns include all supported non-U.S. countries with size/past-return adjustment and equal country weights. Flows are benchmark-relative. Omitted benchmark: Global ex-U.S. Equal appointment weights; hiring-fund-clustered standard errors in parentheses. *, ** and *** denote two-sided significance at 10%, 5% and 1%.

Table 12: Prior records in potential country specialties

Dependent variable	Prior adjusted country return
Units	Annualized basket bps
	(1)
Appointee's education-country link	−218.86**
	(103.19)
Foreign-educated appointee	−55.14
	(35.89)
Log pre-hire destination team size	9.54
	(74.69)
Pre-hire destination unknown education	1035.08***
	(257.23)
Log origin team size	1.43
	(55.61)
Origin-team unknown education	113.97
	(254.52)
Log origin country stock count	−193.92**
	(76.70)
Appointee: top-100 school	−46.52
	(34.20)
Appointee: school ranked 101–500	6.88
	(47.37)
Appointee: female	55.12
	(44.47)
Appointee: unknown gender	−23.44
	(126.33)
Appointee: CFA	−21.10
	(32.48)
Log(1 + manager experience)	57.99
	(36.41)
Prior origin relative NAV (pp)	245.86***
	(22.87)
Continuing-team unknown education	−582.72**
	(251.81)
Hiring-fund × country FE	Yes
Country × appointment-year FE	Yes
Country-appointment observations	9,446
Appointments	1,083
Hiring-fund clusters	257

Notes: Equation (13). The country is held by the hiring fund before appointment and has no continuing manager educated there. The outcome is the recruit's previous teams' adjusted return in that country over four full pre-hire quarters. The education-country link equals one if the recruit studied there. The linked foreign-minus-U.S. comparison is −274.00 (SE 100.61), the sum of the two focal slopes with joint covariance. This model conditions on prior overall NAV performance. Each appointment has equal total weight across supported countries. The 152 linked observations cover 145 appointments at 85 funds. Hiring-fund-clustered standard errors appear in parentheses below coefficients. *, ** and *** denote two-sided significance at 10%, 5% and 1%.

Table 13: Career evaluation and local performance

Dependent variable	Departure	New assignment	Departure	New assignment
Units		Probability in pp		
	(1)	(2)	(3)	(4)
Specification	Baseline	Baseline	With local performance	With local performance
Foreign-educated × fund return (pp)	0.145^{***}	-0.107^{**}	0.138^{***}	-0.108^{**}
	(0.044)	(0.053)	(0.044)	(0.054)
Observed education-country holdings	—	—	−0.17	0.65^{*}
			(0.35)	(0.39)
Local holdings × local return (pp)	—	—	0.017	0.002
			(0.013)	(0.016)
Log recorded fund spell length	2.70^{***}	-2.43^{***}	2.71^{***}	-2.43^{***}
	(0.28)	(0.35)	(0.28)	(0.35)
Female × fund return (pp)	0.071	−0.029	0.071	−0.030
	(0.056)	(0.076)	(0.056)	(0.076)
Unknown gender × fund return (pp)	0.006	0.468^{**}	0.007	0.469^{**}
	(0.139)	(0.210)	(0.139)	(0.210)
CFA × fund return (pp)	−0.050	−0.044	−0.050	−0.044
	(0.043)	(0.050)	(0.043)	(0.050)
Unknown biography × fund return (pp)	−0.107	−0.577	−0.109	−0.580
	(0.248)	(0.380)	(0.249)	(0.380)
Log(1 + experience) × fund return (pp)	0.045	−0.027	0.045	−0.027
	(0.029)	(0.036)	(0.029)	(0.036)
Log(1 + manager experience)	-3.15^{***}	1.74^{***}	-3.15^{***}	1.73^{***}
	(0.41)	(0.56)	(0.41)	(0.56)
Fund × quarter FE	Yes	Yes	Yes	Yes
Fund × manager FE	Yes	Yes	Yes	Yes
Observations	154,677	154,677	154,677	154,677
Fund clusters	1,334	1,334	1,334	1,334

Notes: All columns estimate Equation (14) separately for departures and new assignments. Columns 3–4 add the observed-local-holdings indicator and its interaction with local performance. Outcomes are next-quarter departures and new assignments, in percentage points. Prior-quarter benchmark-relative fund return and local return are in percentage points; interaction coefficients are percentage points per return percentage point. The foreign interaction is a difference in performance slopes relative to U.S.-educated teammates, not an absolute exit probability. Managers receive inverse team-size weights so each fund-quarter has equal total weight. Standard errors, clustered by economic fund, appear in parentheses below coefficients. *, ** and *** denote two-sided significance at 10%, 5% and 1%.

A Descriptive statistics

Tables A1–A4 describe fund mandates, holdings and recorded schooling links. All tests start from the same 1,663 funds; estimation samples differ with outcome availability and the support each comparison requires.

A.1 The within-fund appointment trend

Figure 1 describes the prevalence of foreign-educated managers on current fund teams. Here we examine new appointments. An appointment is a new manager spell at an existing observed team; initial appearances at a fund's launch or the beginning of its history are excluded. The 6,303 known-education appointments include assignments within fund families and moves between firms. The foreign-educated share rises from 32.3% in 2000–2004 to 42.5% in 2020–2024. The supplementary regression distinguishes a rise within funds from changes in which funds appoint managers.

We estimate

$$H_{jq} = \alpha_j + \tau t_q + \Gamma' X_{jq} + e_{jq}. \tag{15}$$

Here H_{jq} is the percentage of known-education appointees who are foreign-educated, measured from 0 to 100. Calendar time t_q is the year of appointment quarter q minus 2012. Controls X contain the size and five traits of the hiring fund's team before the appointments and the unknown-education fraction among new appointees. Observations are fund-quarters with known-education appointments, weighted by their number, with fund effects and fund-clustered errors. The focal coefficient τ is the within-fund appointment-share change per year: +0.60 percentage points per year (+6.0 per decade), significant at 1%, on 4,072 fund-quarters at 1,218 funds (Table A5). Column 2 omits the five team traits on the same observations and estimates +0.57 percentage points per year, also significant at 1%. The within-fund increase is similar with or without those traits.

B Supplementary results

The supplementary tests address three questions behind the main interpretation. First, does the NAV association survive changes in controls and weighting, and can mandate fit or staffing timing explain it? Second, how do portfolio construction, country weights and calendar time affect the local and outside-country findings? Third, can investment opportunities, appointment timing or commercial benefits explain the hiring and career patterns? The appendix follows that order, providing the comparisons and reconciliations supporting the main tests. Panel models use economic-fund clustering; portfolio alphas use the time-series inference in Section 3.4.

B.1 Country relevance measured before appointment

This extension of Equation (3) asks whether a poor match between managers' education countries and the fund's mandate explains the shortfall. If foreign-educated managers often run mandates in which their education countries carry little weight, the association should be more favorable where those countries matter more to the benchmark. We therefore test whether it is more favorable when managers' schooling countries have greater benchmark relevance at appointment. Let a_{ih} denote benchmark capital in manager i's distinct non-U.S. education countries, averaged over the four quarter-ends strictly before appointment h. Schooling includes undergraduate and graduate degrees. The denominator is total benchmark capital, including U.S. equities, and multiple degrees in the same country count once. All four quarters must have at least 95% of benchmark weight mapped to countries; a 105% upper bound allows rounding. Missing pre-appointment coverage is classified as unknown. The recorded benchmark identifier is static, so a_{ih} uses historical weights of that index rather than a verified historical mandate.

The added regressor Relevance_{jq} is the sum of measured a_{ih} across current foreign-educated managers divided by the number of current managers with known education; unmeasured contributions are set to zero and identified separately by $\text{UnknownRelevance}_{jq}$, the share foreign-educated with unknown relevance. We add $\theta\text{Relevance}_{j,q-1}$ and $\eta\text{UnknownRelevance}_{j,q-1}$ as separate terms to Equation (3), with its existing

ρForeignShare$_{j,q-1}$ retained. Unknown education remains a distinct control. Relevance is fixed within each appointment spell; changes in the current team's membership and composition generate variation within funds. No fund-month is dropped for missing relevance, which affects 50.4% of foreign manager-quarter observations.

Table A6, column 2, also includes foreign education-country shares. A manager with K distinct foreign education countries contributes $1/K$ to each, aggregated over the same team denominator. Their sum is ForeignShare, so the UK share is omitted; Lebanon is absorbed by the fixed effects. Accordingly, ρ in this column is conditional on the omitted country and zero measured relevance, not the main paper's average foreign-share comparison. The coefficient of interest is θ, which asks whether a given foreign-educated share performs better when its schooling is more relevant to the mandate. The additional controls separate differences in educational geography from relevance: for example, the team share educated in Japan is distinct from how important Japan was to the benchmark before those managers joined. The change from column 1 to column 2 therefore reflects conditioning on educational geography, rather than a new sample or risk model. The positive foreign-share coefficients in these augmented models describe the reference point of zero measured relevance; they are not the average share comparison in Table 2. The raw units of θ are annual NAV basis points per unit of team relevance. A unit change would move a fully foreign-educated team from education countries with no benchmark weight to countries that make up the entire benchmark, so we report the interquartile contrast. The interquartile contrast is 0.1050θ for a fully foreign-educated team with known relevance: -48.6 in column 1 and -72.3 in column 2, with 95% intervals $[-109.4, 12.2]$ and $[-122.7, -21.9]$. At a smaller fixed foreign share, the contrast scales with that share if known-relevance managers' exposure changes by the same amount. Using current benchmark weights instead of pre-appointment weights gives imprecise gradients with and without country-composition controls; that alternative also changes which managers have measured relevance. The adverse association reported here concerns relevance recorded at appointment. The contrast compares foreign-educated teams with one another and does not add to the -70-basis-point gap in Table 2: returns are weaker, not stronger, where managers' education countries matter more to the mandate.

B.2 Precision weighting

Table 2, column 4 asks whether giving more weight to funds with less noisy returns changes the estimated performance gap. It retains the factor-adjusted net NAV outcome, observations, controls and fixed effects of Table 2, column 1, and weights observations by the inverse of the fund's residual return variance. The foreign-education coefficient is -69.7 annual basis points (standard error 22.7), close to the equally weighted estimate. The similar coefficients show that giving more weight to less noisy returns changes precision more than the estimated magnitude.

B.3 Further net-return specifications

We check whether the NAV association depends on the treatment of prior flows, measured portfolio composition or extreme returns. Quarterly flow is $\text{Flow}_{jq} = 100[\text{TNA}_{jq} - \text{TNA}_{j,q-1}(1+R^{\text{net}}_{jq})]/\text{TNA}_{j,q-1}$, where TNA is quarter-end assets and R^{net}_{jq} the compounded quarterly net return in decimals. Thus Flow is in percentage points of beginning-quarter assets; the NAV model uses the preceding quarter's $\text{Flow}_{j,q-1}$ for the months of quarter q. On the identical 206,847 fund-months, replacing raw prior-quarter flow with $\text{asinh}(\text{Flow}_{j,q-1}/10)$ changes the foreign-education coefficient from -69.51 to -69.49 annual basis points. The paired raw-minus-transformed difference is -0.02 basis points, with a 95% interval of $[-0.45, 0.42]$. We use raw flow for its simpler interpretation. Both comparisons retain uncapped flow observations; the exercise tests the sensitivity of the education coefficient to the control's transformation.

Table A7 retains Equation (3), its sample and fixed effects. Column 1 adds portfolio size and momentum tilts, the holdings-weighted average size and past-return terciles (coded 0 to 2) of the fund's foreign stocks within the country groups that define the stock benchmarks in Section 2.4, to the model without expenses, turnover and concentration; column 2 instead adds top-ten concentration and turnover. Missing-value indicators keep observations whose tilts, concentration or turnover are unavailable. Column 3 uses the full controls and winsorizes the risk-adjusted outcome at the 1st and 99th percentiles within benchmark-month

cells. All use equal fund-month weights and economic-fund clustering. The focal education-share coefficients are −66, −69 and −69 annual basis points. These comparisons assess the stability of the return gap to measured portfolio composition and extreme outcomes.

B.4 Staffing timing and selection sensitivity

Two diagnostics examine staffing timing and selection on observed characteristics. Funds do hire country specialists after local losses (Section 4.1); the first diagnostic asks whether such staffing responses could produce the fund-level coefficient. If weak current returns forecast a future shift toward foreign-educated managers, the current share could stand in for that later response because team composition is persistent; future team composition would then help explain the current association. We add future foreign-educated share and future unknown-education share to Equation (3), measured four quarters after the team in Equation (3): $\rho_{+4}\text{ForeignShare}_{j,q+3}$ and $\xi_{+4}\text{UnknownShare}_{j,q+3}$, where q is again the quarter containing month m.

The controls X are those of Equation (3), including the current unknown-education share. The coefficient ρ_{+4} measures the additional predictive content of future foreign-educated share, conditional on the current team and controls. On the 193,986 fund-months with teams observed four quarters ahead, the current-share coefficient is −75 basis points (Table A8, column 1). Including future foreign-educated and unknown-education shares leaves it at −72; the future foreign-educated share has a coefficient of −3 with a standard error of 37 (column 2). The current association is thus stable to future team composition, which adds no clear independent predictive relation.

The second diagnostic asks whether observed fund characteristics account for the negative coefficient. With only the unknown-education fraction controlled, the estimate is −59 basis points; adding the other controls makes it −70 (column 3 and Table 2). Within the proportional-selection calibration of Oster (2019), the adjustment at equal observed and unobserved selection gives −73 basis points, while the selection parameter needed for a zero coefficient is −20.6.[30] The observed-control adjustment moves the coefficient away from zero. Both checks show a coefficient that is stable to the staffing and fund conditions measured in Equation (3).

B.5 Performance beyond the initial appointment period

To test whether the NAV association is confined to newly appointed managers, Table A9 modifies Equation (3) by replacing $\text{ForeignShare}_{j,q-1}$ with three shares: $\text{ForeignShare}^{\text{new}}_{j,q-1}$, $\text{ForeignShare}^{\text{est}}_{j,q-1}$ and $\text{USShare}^{\text{new}}_{j,q-1}$. Their coefficients are ρ_F^{new}, ρ_F^{est} and ρ_{US}^{new}, respectively.

The three shares count new foreign-educated managers, established foreign-educated managers and new U.S.-educated managers, respectively, divided by all managers with known education. Established U.S.-educated managers form the omitted category. New means within the first two years (eight quarters) at the fund, and established means later; as in the NAV equation, q contains the return month and the shares are measured at quarter-end $q-1$. The reduced controls X contain unknown schooling, team and asset scale with spline terms, and the five team traits. Fund and benchmark-month effects, the predetermined thirteen-factor outcome and fund clustering are retained; OLS and precision-weighted columns use the same model. The focal established-share coefficient ρ_F^{est} is −81 annual basis points in OLS and −82 in WLS, significant at 5% and 1%. The new foreign-minus-U.S. contrast, $\rho_F^{\text{new}} - \rho_{US}^{\text{new}}$, is −35 in OLS and imprecise. The NAV association is present among teams with established foreign-educated managers. This comparison distinguishes teams by current managers' tenure; the appointment analysis separately tracks local performance after a hire.

[30] We set $R_{\max} = 1.3$ times the within-fund R^2 of the controlled regression. Within explanatory power is low for monthly returns, so the calibration is driven substantially by the direction of coefficient movement. It is a sensitivity calculation under proportional selection, not a measured amount of unobserved confounding.

B.6 Alternative local-portfolio constructions

Table A10 asks whether the high-conviction result depends on how conviction is measured, which holdings and education links enter, or the risk model. Each column changes one element of the baseline and re-estimates the top-portfolio alpha with Equation (8).

The first two columns change the conviction measure. The baseline measure in Equation (7) subtracts each held stock's share of the market capitalization of the fund's holdings in the country. Column 1, the benchmark reference, uses the assigned benchmark's weights instead, renormalized over the same held stocks:

$$I^B_{jkq} = \frac{\omega_{jkq}}{\sum_{k' \in \Omega_{jcq}} \omega_{jk'q}} - \frac{b_{jkq}}{\sum_{k' \in \Omega_{jcq}} b_{jk'q}},$$

where b_{jkq} is stock k's weight in fund j's assigned benchmark at quarter-end q. When the benchmark holds none of the fund's stocks in the country, the second term is dropped and positions are ranked by their weights within the country. Column 2, the two-year reference, keeps capitalization weights but widens the reference set from the stocks held at q to Ω^8_{jcq}, all stocks the fund held in the country at any quarter-end from $q-8$ through q:

$$I^U_{jkq} = \frac{\omega_{jkq}}{\sum_{k' \in \Omega^8_{jcq}} \omega_{jk'q}} - \frac{\mathrm{MC}_{kq}}{\sum_{k' \in \Omega^8_{jcq}} \mathrm{MC}_{k'q}}.$$

In both columns, the top quartile of the alternative measure within each fund-country-quarter defines the high-conviction positions.

The remaining columns keep the baseline measure. Column 3 removes U.K. equities from portfolio construction, because the United Kingdom is the most common foreign education country. Column 4 defines education countries by undergraduate degrees only. Columns 5 and 6 add developed and emerging momentum factors, retaining or omitting the common portfolio benchmark; column 7 retains only that benchmark; and column 8 deducts estimated bid-ask trading costs. Each column reports one top-portfolio regression with all loadings; the table notes report the paired Wald tests of the top-minus-ordinary alpha. The baseline is shown only in Table 6.

Alphas range from 2.75% to 5.74%. The top-minus-ordinary spread remains significant at 5% in columns 3 through 8. It is marginal with the two-year reference ($p = 0.057$) and imprecise with the benchmark reference ($p = 0.251$).

B.7 Contribution of local positions and country weighting

This appendix measures what education-country positions add to the funds' returns and explains why that contribution is much smaller than the strategy alphas in Section 3.4 would suggest. We measure the contribution per dollar of pooled covered equity capital: the value of sample funds' disclosed stocks with usable returns, including U.S. stocks. Let s^L_m be the education-country share of covered equity capital in month m and R^L_m the return on those holdings at actual capital weights. Their monthly contribution to the covered equity return is

$$C^L_m = s^L_m R^L_m, \qquad R^{\text{all}}_m = C^L_m + C^O_m. \tag{16}$$

Here R^{all}_m is the return on the entire covered equity portfolio and C^O_m the contribution of the other covered equities. The annualized raw contribution is the average of C^L_m over the 300 months, times twelve, expressed in basis points. Weighting each month's return by that month's capital share captures comovement between capital shares and returns, which multiplying the average share by the average return would miss. To measure the contribution beyond risk exposures, we use the excess-return contribution $C^{\text{ex}}_{pm} = s_{pm}(R_{pm} - R^f_m)$ as the dependent variable in Equation (8), where s_{pm} is the share of covered equity capital in group p, all local, top local or ordinary local positions, and R_{pm} is now the group's return at actual capital weights. The intercept is then the alpha these positions add per dollar of pooled covered equity, rather than per dollar invested in them (Table A11).

Education-country holdings represent 8.7% of covered capital. They contribute 57.5 basis points a year of raw equity return but no clearly positive alpha: their alpha contribution is −10 basis points (standard error 11). The high-conviction positions occupy 3.6% of covered capital, and their alpha contribution is

0.3 basis points a year (standard error 6.1), well below the 14 basis points implied by applying their 4.0% strategy alpha to that share.

The difference comes from country weights. The 4.0% strategy alpha holds the selected stocks in benchmark country proportions. Table A12 holds the fund-country-month cells fixed while comparing top, ordinary and combined local portfolios under benchmark and actual country weights. On this common sample, the top local portfolio earns 407 basis points a year under benchmark country weights but 17 (standard error 159) at funds' actual country weights. The strategy alpha thus measures the value of the stock choices; the contribution measures how much of that value reaches the portfolio at the allocations funds choose. These comparisons concern the constructed local strategies; the joint country panel in Table 5 separately compares teams with U.S.-educated peers.

Table A12 re-estimates Equation (8) for six portfolios built from identical fund-country-month cells. Columns 1–3 use benchmark country shares for top, ordinary and combined local holdings; columns 4–6 use actual capital weights. Risk factors, the common portfolio benchmark and time-series inference are those of Table 6. Because the stock baskets are the same, the comparison changes only the capital assigned to their countries. On 77,188 identical fund-country-month cells at 749 fund identifiers, actual country weights lower the combined local alpha by 234 annual basis points relative to benchmark weights, significant at 1% ($p = 0.007$); the table notes report this difference, $12 \times 10^4(\alpha_{\text{actual,all}} - \alpha_{\text{benchmark,all}})$. The combined portfolios are estimated directly because the capital shares of top and ordinary holdings change over time. The comparison is conditional on the local book: it is not a 234-basis-point loss on total fund capital and cannot be added to the panel's education coefficient.

B.8 Changes in the geographic performance gap

We next ask whether the foreign-versus-U.S. comparison in outside countries changes over the twenty-five-year sample. To make the local and outside comparisons consistent, we retain the full country panel in Equation (6) and allow both education associations to vary with calendar time:

$$\begin{aligned} R_{jcq} = \alpha_{jc} + \lambda_{cq} + \beta \mathbb{1}_{jcq}^{\text{Link}} + \gamma \text{ForeignShare}_{jq} \\ + \beta^T \mathbb{1}_{jcq}^{\text{Link}} \times t_{q+1} + \gamma^T \text{ForeignShare}_{jq} \times t_{q+1} + \Gamma' X_{jcq} + \varepsilon_{jcq}. \end{aligned} \tag{17}$$

As in Equation (15), t_q is the year of quarter q minus 2012, in single years; the superscript T marks the time-interaction coefficients. Holdings and education at quarter-end q predict returns in $q+1$, so the interactions use the payoff year t_{q+1}. The sample, controls, weights, fixed effects and clustering are those of the joint country model in Section 3.3.

The outside-country gap at date t is $g_{\text{out}}(t) = \gamma + \gamma^T t$. For a fully foreign-educated team in an education country, the comparison with a U.S.-educated team is $g_{\text{local}}(t) = \beta + \gamma + \beta^T t + \gamma^T t$. Each uncertainty calculation uses the joint covariance of its components. The endpoint gaps are fitted values from the full-sample trend regression.

The outside-country gap becomes more negative over time. In Table A13, column 1, γ^T is -7.06 annualized basket basis points per calendar year, significant at 5%. The fitted gap reaches -124.5 by 2024, with a standard error of 48.3. Inverse-residual-variance weighting gives a similar deterioration and endpoint gap (column 2). Figure A1 plots the outside and local comparisons from these same regressions. The late-sample outside shortfall is about 1.25 percentage points per dollar of the country basket.

The education-country comparison remains imprecise in 2024 under both weighting schemes. The evidence for local expertise comes from the direct conviction-portfolio contrasts in Section 3.4.

Country-quarter effects remove common national-market returns, and the stock comparisons control size and momentum. The trend describes how the team-composition association changes over the sample period. It complements the familiarity tests by showing when the broader outside-country gap is most pronounced; it does not track the performance of individual managers as they gain tenure.

B.9 Familiarity results with raw country returns

The adjusted country return in Equation (2) subtracts the return on a matched portfolio of local stocks with similar size and past returns. Table A14 re-estimates Equation (9) with the raw return on the fund's holdings in each country instead. The observations, controls, fixed effects, weights and clustering are those of Table 7. Country-quarter effects remain, so the comparison is still among funds holding the same country in the same quarter; only the matching on stock size and past returns is dropped.

The familiarity pattern remains. The difference between the foreign- and U.S.-educated slopes, θ, is -346 basis points for a shared border (standard error 155, significant at 5%), -195 for a common official language (standard error 69, significant at 1%) and -81 basis points per log point of social connectedness (standard error 41, significant at 5%). The foreign-educated slopes, $\delta + \theta$, are -157 for borders (standard error 74, significant at 5%), -62 for official language (standard error 64) and -35 per log point of social connectedness (standard error 17, significant at 5%). The raw differences are 15 to 30 percent smaller than those in Table 7. For borders, the change comes mainly from U.S.-educated teams' holdings in Canada and Mexico, whose size and past-return characteristics earned lower returns: without the benchmark, the U.S.-educated border slope falls from +300 to +188 basis points, while the foreign-educated slope moves only from -184 to -157.

B.10 Performance following country-linked appointments

Section 4.1 assesses the payoff to seeking local expertise after disappointing performance. Starting with the fund-country-quarters of Equation (10), we retain appointments of managers educated in the held country (Table 8, column 1) through 2023Q4, allowing four subsequent payoff quarters through 2024Q4. A disappointing prior record means a negative twelve-month country-market return or a negative own country-relative return. Each appointment is matched to up to three other economic funds sharing the investment country, assigned benchmark, appointment quarter and disappointing-performance status, with no appointment that quarter. The closest controls minimize standardized distance in prior country-relative returns, log country capital, log stock count and incumbent schooling coverage. Matching uses only information available before appointment; controls may already employ foreign-educated managers and may hire later.

Equation (11) compares changes within each fund-case with changes at its matched funds in the same payoff quarter. We retain quarters -4 through -1 and $+1$ through $+4$, requiring at least three jointly observed quarters on each side and omitting quarter zero. The common panel contains 13,237 observations in 511 cases, 267 hiring funds and 610 economic-fund clusters. Matching is unchanged when we require common support for the three outcomes, and unsupported controls are not replaced. Each hiring fund-quarter has equal total weight, divided across its country cases, hiring and comparison sides, pre- and post-appointment phases, and observed quarters within each phase. The controls are those of the main NAV model; its foreign-share regressor is replaced by the hiring-by-post interaction. The concentration-missingness indicator is zero throughout this supported panel and is absorbed. Fund clustering accounts for repeated hiring and comparison roles.

The local outcome follows the characteristic-adjusted country-return construction in Section 2.4. The outside outcome pools the corresponding adjusted country returns at their monthly capital weights. For each case, the education geography consists of the non-U.S. schooling countries of the hiring team's current managers and dated appointees in the appointment quarter. We hold this set fixed for all quarters and apply it to both sides of the comparison, so appointment does not mechanically reclassify holdings. Outside countries here are outside that common set, rather than outside each comparison fund's own schooling geography. NAV uses the factor-adjusted monthly return of Equation (1), averaged within the actual payoff quarter. All outcomes use the return construction of the main tests; complete stock-return coverage is not required and missing country baskets are not assigned zero returns.

Table 9 reports one regression per outcome on exactly these observations. The local coefficient is +228 annualized basket basis points; the outside-country and NAV coefficients are -15 and +37, respectively. The local estimate is significant at 1%; the other two estimates do not establish an outside offset or a whole-fund gain. The basket coefficients have different capital denominators and cannot be added to obtain the NAV coefficient. The purpose of the common design is to compare the scope of improvement after a targeted appointment, rather than to treat the full-period team-composition coefficient as its hiring effect.

The appointments are observational and the controls evolve during the window, so these are conditional relative changes.

To assess persistence, Table 10 extends the same matches through eight subsequent quarters. Appointments must occur by 2022Q4 to allow two full years within the sample period. We require at least three jointly observed local-return and NAV quarters in each of three phases: quarters -4 to -1, $+1$ to $+4$, and $+5$ to $+8$, omitting quarter zero. The common cohort contains 430 country-appointment cases at 231 hiring funds, 1,339 fund-case units, 15,862 observations and 514 economic-fund clusters. We keep the matches of the one-year comparison, without requiring that recruits stay or that comparison funds refrain from later hiring. The cohort is smaller than the first-year comparison because of the longer calendar and outcome-support requirements.

We replace $\beta_H \mathbb{1}_{ja}^{\text{Hire}} \times \mathbb{1}_{at}^{\text{Post}}$ in Equation (11) with $\beta_{H,1} \mathbb{1}_{ja}^{\text{Hire}} \times \mathbb{1}_{at}^{\text{Year1}} + \beta_{H,2} \mathbb{1}_{ja}^{\text{Hire}} \times \mathbb{1}_{at}^{\text{Year2}}$. The year indicators select quarters $+1$ to $+4$ and $+5$ to $+8$, respectively; both coefficients measure improvement relative to the pre-appointment year and to comparison funds. Year two is the average return during months 13–24, not the cumulative two-year return. The controls and fixed effects are unchanged, and equal hiring-fund-quarter weights are divided across cases, hiring/comparison sides, the three phases and observed quarters. The contrast $\beta_{H,2} - \beta_{H,1}$ uses their joint fund-clustered covariance. The estimated local decline is -205.34 basis points (standard error 96.72, $p = 0.034$). The NAV contrast is $+26.03$ (standard error 34.02), so the local fading pattern does not establish a decline in whole-fund performance.

B.11 Timing of the full-sample NAV association

To relate the appointment comparison to the main NAV result, Table A15 locates the observations that contribute to the full-sample coefficient. We retain all 206,847 fund-months, the original controls, fund effects and benchmark-month effects. After removing those controls and effects from both foreign share and factor-adjusted NAV, each observation contributes its residual share times its residual return, divided by the full-sample sum of squared residual shares. Summing these terms reproduces the -69.5-basis-point coefficient exactly. Grouping the terms by appointment timing is descriptive accounting, not a set of subgroup regressions or hiring effects.

The 511 country cases identify 460 distinct appointment quarters at 267 hiring funds; we call these the studied appointments. We classify their monthly returns as before the first studied appointment, in an appointment quarter, in the following year, in years two through five, or later; the clock resets after another studied appointment. Funds without a studied appointment form the remaining group. The first studied appointment is not necessarily the fund's first foreign-educated hire: foreign-educated managers already represent 41.7% of teams, on average, in the pre-appointment observations. Those observations contribute -37.1 basis points to the -69.5 coefficient. The full-history association therefore substantially predates the targeted appointments whose local payoff we estimate. The table clarifies how a positive local appointment comparison can coexist with a negative broad NAV association, without assigning the latter to the appointments.

B.12 Performance-conditioned monthly flows

A commercial payoff could make the background valuable even without stronger investment performance; investor preferences can shape fund incentives (Gantchev et al., 2024). The broad monthly flow test separates a potential commercial value of foreign education from investors' responses to prior returns. Let TNA_{jm} be end-of-month assets and R_{jm}^{net} the monthly net return in decimal units. Flow in percentage points is $\text{Flow}_{jm} = 100[\text{TNA}_{jm} - \text{TNA}_{j,m-1}(1 + R_{jm}^{\text{net}})]/\text{TNA}_{j,m-1}$. Before estimation, this monthly flow measure is winsorized at the pooled 1st and 99th percentiles; the cutoffs appear in Table A16. This treatment applies to the monthly flow outcome, not the raw quarterly flow control in the NAV model. We estimate

$$\begin{aligned} \text{Flow}_{jm} = \alpha_j + \lambda_{bm} + \delta \text{ForeignShare}_{j,q-1} \\ + \Gamma_S' S_{j,m-1} + \Gamma_X' X_{j,m-1} + \Gamma_Z' Z_{j,q-1} + u_{jm}. \end{aligned} \tag{18}$$

$\text{Rank}_{j,m-1}$ is the zero-to-one percentile rank of the twelve-month compounded return ending in $m-1$ among same-benchmark funds. Suppressing those subscripts within the three expressions, $S_{j,m-1}$ contains $\min(\text{Rank}, .2)$, $\min(\max(\text{Rank}-.2, 0), .6)$ and $\max(\text{Rank}-.8, 0)$. These terms allow flows to respond differently to poor, middle and strong prior performance. X contains log lagged assets, log fund age, the standard deviation of the preceding twelve monthly returns, lagged annual expenses and log(1+turnover). Z contains log team size, unknown-education share and the five biography traits. As in Equation (3), b is the assigned benchmark and team measures use the quarter-end $q-1$ preceding the quarter q that contains month m. The observed-flow sample has 185,748 months at 1,494 funds; missing required characteristics are not filled with zero.

The coefficient δ asks whether foreign-educated representation predicts additional flows among funds with comparable prior performance and characteristics. Table A16 gives $\delta = 0.207$ monthly percentage points for the full-share difference, with a standard error of 0.153 and a 95% interval from -0.093 to $+0.508$. A positive commercial premium is possible but not precisely established. Column 2 omits expenses and turnover on the same observations; the foreign-share coefficient remains 0.21 percentage points (standard error 0.15), so conditioning on these operating choices does not explain the imprecision. The quarterly and appointment comparisons in Table A17 use their own outcome definitions and samples. With the prior-flow record in Section 4.2, these tests cover fundraising both before and after appointment.

B.13 Commercial outcomes at quarterly and appointment horizons

Table A17 asks whether the qualification brings a commercial payoff despite its weaker investment-return association. The outcomes $Y_{jq}^{(v)}$ pair the team observed at quarter-end q with subsequent flows and reported fee rates: flow in quarter $q+1$ (columns 1–2), the sum of flows over quarters $q+1$ to $q+4$ (column 3), and the expense ratio reported for the previous year (column 5). Flow is measured in percentage points of beginning-quarter assets and expenses in annual basis points. For each outcome v, we estimate

$$Y_{jq}^{(v)} = \alpha_j + \lambda_{bq} + \delta_v \text{ForeignShare}_{jq} + \Gamma_v' X_{jq} + u_{jq}^{(v)}. \tag{19}$$

Fund effects α_j and benchmark-quarter effects λ_{bq} follow Equation (3). The focal coefficient δ_v compares an entirely foreign-educated team with an entirely U.S.-educated team, conditional on the stated controls. A positive flow coefficient would indicate greater subsequent asset gathering; a positive expense coefficient would indicate a higher reported fee rate. The latter is a comparison of existing fees, not a subsequent fee change. The controls X contain UnknownShare, log team size and lagged assets with their spline terms, and the five team traits. Column 2 adds R_{jq}^{past} and the interaction $\text{ForeignShare}_{jq} \times R_{jq}^{\text{past}}$ as separate regressors, where R_{jq}^{past} is the prior-year relative NAV return, the same variable as the NAV-model control, in annual percentage points. It differs from the single-quarter career signal R^{rel}. Its interaction measures the foreign-minus-U.S. difference in flow-performance sensitivity.

Column 4 asks the more specific question of whether foreign-educated appointments are followed by stronger fundraising. Its outcome is the sum of flows over the four quarters after appointment. In Equation (19), it replaces current-team ForeignShare_{jq} with the appointee fraction s_{jq}^H and benchmark-quarter effects with appointment-year effects; we denote the appointee-share coefficient by δ_H.

The fraction s_{jq}^H is measured among known-education appointees. It represents the same composition as H_{jq} in Equation (15), expressed from zero to one rather than zero to 100. Thus δ_H compares subsequent four-quarter flows after all-foreign versus all-U.S. appointments, conditional on prior fund conditions. The controls X contain pre-appointment four-quarter flows, the prior-year relative NAV return, lagged assets, the size and traits of the hiring fund's team before the appointment, and unknown education among new appointees. All commercial models use equal observation weights and cluster by economic fund. The full-share differences are -0.29 flow percentage points next quarter, -0.26 over four quarters, and $+0.22$ over four quarters after appointment. The expense difference is -0.39 annual basis points. These estimates do not establish a commercial premium. The flow-performance interaction is -0.04 percentage points per

annual NAV percentage point (standard error 0.06), also imprecise; it asks whether investors' flow response to performance differs with the team's education share.

B.14 Prior records within origin peers and hiring rounds

The main hiring comparison controls for the benchmarks of recruits' previous funds. The checks here ask whether their record gap persists after a finer comparison with other funds facing the same investment opportunities, or after comparing only recruits appointed together. Both use outside-country records; the main comparison also includes education countries. For each origin-country-quarter record, we subtract the mean selection of other economic funds sharing its assigned benchmark, investment country and quarter; all share classes of the focal economic origin are excluded. Peer share classes are combined before averaging economic funds. We then apply the same equal-country, equal-quarter and equal-economic-origin record weights to form $\bar{R}_h^{\text{peer}}$, which replaces $\bar{R}_h^{\text{pre}}$ as the outcome in Equation (12).

The outside-country record and its peer-adjusted counterpart use identical supported countries, 1,321 appointments and 27 regressors. The main all-country record adds supported education-country holdings for those same appointments. Both retain hiring-fund and year effects, equal appointment weights and fund clustering; neither controls for prior NAV. Benchmarks are supplied assigned mandates, rather than independently reconstructed historical mandate changes.

The appointee controls in both columns of Table A18 are those of Equation (12); the school-rank indicators mark degrees from the top 100 and from ranks 101–500 of the 2003 Academic Ranking of World Universities, with other schools omitted. The peer-adjusted foreign-education coefficient is −71.8 annualized basket basis points, with a standard error of 41.0, significant at 10% (Table A18, column 1). The gap remains negative after comparing the previous teams with funds facing the same country, quarter and benchmark conditions. The corresponding outside-country estimate without peer subtraction is −100.8; the all-country main estimate is −86.4. Comparing the three estimates separates the effect of peer adjustment from that of country scope.

A separate comparison asks whether the record gap remains among recruits accepted together. Table A18, column 2 uses the outside-country version of $\bar{R}_h^{\text{pre}}$ as the outcome in Equation (12), replaces separate fund and year effects with hiring-fund-by-appointment-quarter effects α_{jq}, where q is the appointment quarter, and omits $\mathbf{B}_h$. We denote its foreign-education coefficient by δ_{round}.

Hiring-fund-by-appointment-quarter effects hold common recruitment conditions fixed; destination team size and unknown schooling are absorbed. This model uses all available outside-country records and no origin benchmark or prior NAV control. Equal appointment weights and hiring-fund clustering are retained. On 832 appointees, δ_{round} is −81 annualized basket basis points, with a standard error of 41 ($p = 0.053$). The negative gap remains among appointees accepted together, providing a separate check on recruitment conditions from the origin-mandate adjustments above.

B.15 Country coverage and career evaluation

A fund may retain a specialist because colleagues cannot replace the country knowledge. We ask whether the relative departure response is concentrated among managers whose recorded schooling link no colleague supplies. Table A19 separates the career comparison by schooling coverage: a foreign-educated manager has unique coverage if at least one held education country lacks another current manager's schooling link. Coverage is duplicated when the manager matches held countries but all those links are also supplied by colleagues; it is unmatched when no education country is held and unclassified when prior holdings are missing. These categories are measured each quarter from the previous quarter's holdings and the current team. The equation replaces $\mathbb{1}_i^{\text{Foreign}} \times R_{jq}^{\text{rel}}$ in Equation (14) with category interactions:

$$100 \times \mathbb{1}_{ijq}^{\text{Exit}} = \mu_{jq} + \nu_{ij} + \sum_g \kappa_g \mathbb{1}_{ijq}^{g} \times R_{jq}^{\text{rel}} + \sum_{g \neq \text{NoMatch}} \eta_g \mathbb{1}_{ijq}^{g} + \Gamma' X_{ijq} + e_{ijq}. \tag{20}$$

Here g indexes unique, duplicated, no-match and unclassified coverage, and $\mathbb{1}_{ijq}^{g}$ is the corresponding foreign-educated-manager indicator; U.S.-educated colleagues are the comparison. Because the four indicators sum

to $\mathbb{1}_i^{\text{Foreign}}$, which the fund-manager effects absorb, the no-match level is omitted, and all four performance interactions enter alongside the career controls, fund-quarter and fund-manager effects, inverse team-size weights and fund clustering. The focal κ_g coefficients are relative exit-response slopes per percentage point of better performance: +0.180 for unique coverage, +0.006 for duplicated coverage and +0.122 for no held-country match. Multiplying by a ten-point deterioration gives relative departure differences of −1.80, −0.06 and −1.22 percentage points. Unique and no-match slopes individually exclude zero, but direct group differences are imprecise; the joint zero-response test for duplicated and no-match coverage has $p = 0.141$. Relative protection can therefore appear beyond currently matched holdings without precisely distinguishing the coverage groups. Coverage measures recorded schooling in held countries, rather than verified responsibilities or future country plans. Column 2 removes the other observed traits' performance interactions on the identical sample. The unique-coverage slope remains +0.177 (standard error 0.056), significant at 1%; the no-match slope is +0.113 (standard error 0.061), significant at 10%. The similar points preserve the pattern, while the no-match comparison is less precisely estimated.

C Variable Definitions and Timing

Table A20 defines the main variables. A basis point is one hundredth of a percentage point. Coefficients on ForeignShare compare foreign-educated shares of one and zero among managers with known schooling, holding the unknown-education fraction fixed. Fund returns, adjusted country returns and portfolio alphas are all annualized but have different capital denominators and cannot be added. Missing education is coded as unknown, and an unavailable return is not treated as zero.

Fractions are coded from zero to one unless a percentage-point unit is stated. Return controls and signals use percentage points; capital coverage uses percentage points; calendar interactions use years from the stated reference date. Social connectedness enters in log units without standardization. Economic changes such as ten percentage points or one standard deviation are calculated in the text. Log experience is $\log(1 + \text{years of experience})$; the team measure is its mean. The two nonlinear asset terms are restricted-cubic-spline bases in log lagged assets, with knots $(0.5921, 4.1586, 5.8772, 8.4417)$; the corresponding team-size knots are $\log(1, 2, 3, 5)$. The linear log term is reported separately. The prior relative NAV control in the NAV and quarterly commercial models is the preceding year's annualized monthly mean in percentage points; the monthly flow model instead uses ranks of compounded twelve-month returns. Prior relative volatility is annualized in percentage points before applying $\log(1 + \cdot)$. The main and peer-adjusted appointment selection models contain origin benchmark indicators and no prior NAV control. The appointment-flow and education-country prior-record models retain a mean pre-appointment quarterly benchmark-relative NAV signal in percentage points; its payoff quarters run from five to two quarters before the appointment quarter. Prior flow enters directly in percentage points of beginning-of-quarter assets, with a missing-information indicator. Separate missing-information indicators retain unavailable controls. Fixed-effect intercepts and regressors absorbed by them are identified in the table notes rather than assigned estimated zero coefficients. The predetermined factor exposures used to construct NAV risk-adjusted returns are outcome inputs, not common second-stage slope regressors. The portfolio-alpha tables instead report every estimated factor loading, in dimensionless units, alongside annualized alpha.

Table A1: Fund assets by assigned benchmark

Morningstar Index	Mean assets (USD billions)	Mean active funds
Morningstar Gbl NR USD	4.672	51.55
Morningstar Gbl xUS Val TME NR USD	2.853	86.63
Morningstar Gbl Growth TME NR USD	2.790	51.36
Morningstar Gbl xUS Growth TME NR USD	1.787	86.45
Morningstar Gbl xUS NR USD	1.260	164.27
Morningstar EM Americas NR USD	1.091	2.57
Morningstar DM APAC xJpn NR USD	0.800	14.87
Morningstar EM NR USD	0.757	125.59
Morningstar Gbl Val TME NR USD	0.606	38.64
Morningstar Gbl xUS SMID NR USD	0.550	56.65
Morningstar DM Eur NR USD	0.528	10.16
Morningstar Gbl SMID NR USD	0.460	18.04
Morningstar Jpn NR USD	0.367	6.99
Morningstar DM APAC NR USD	0.257	3.15
Morningstar China NR USD	0.180	13.05
Morningstar India NR USD	0.121	2.34
Morningstar Gbl Lrg NR USD	0.093	28.97

Notes: This table reports the mean fund-level total net assets and the mean number of funds in each Morningstar benchmark category, computed at quarterly frequency over the 1,663-fund sample. Because the counts are per-quarter averages, the "Mean active funds" column reflects funds active in a given quarter and sums to about 760—the average number of funds active per quarter—rather than the 1,663 distinct funds observed over the full sample, since funds enter and exit over time. Index names follow Morningstar: Gbl is global, xUS excludes the United States, DM and EM are developed and emerging markets, APAC is Asia-Pacific, xJpn excludes Japan, Lrg is large-cap, SMID is small- and mid-cap, Val is value, TME denotes Morningstar's target-market-exposure indexes, and NR USD denotes net returns in U.S. dollars.

Table A2: Portfolio holdings

	Pooled over all observations			
	Mean	Median	Std. dev.	N
Sample Funds				
# of countries held	18.99	19	8.44	67,380
Country-level portfolio share(%)	4.83	2.17	9.47	1,279,345
# of stocks held	210.26	81	589.15	71,742
Stock-level portfolio share(%)	0.45	0.09	0.88	15,084,718
Morningstar Gbl xUS NR USD				
# of countries held	49.47	51	3.46	88
Country-level portfolio share(%)	1.97	0.58	3.35	4,353
# of stocks held	5,136.20	5,104	982.78	88
Stock-level portfolio share(%)	0.02	0.00	0.06	451,986
All Benchmarks (equal weights)				
# of countries held	27.55	26	20.28	1,947
Country-level portfolio share(%)	3.43	0.57	10.15	53,639
# of stocks held	1,887.58	1,344	1,841.74	1,947
Stock-level portfolio share(%)	0.05	0.01	0.23	3,675,116

Notes: This table reports country-level and stock-level holding counts and portfolio shares for the sample funds, the Morningstar Global ex-US benchmark, and all Morningstar benchmarks assigned to sample funds. The sample-fund rows are computed on the same 1,663-fund sample and 2000–2024 window as Table 1. Each statistic pools all observations counted in N: the rows counting countries or stocks held pool fund-quarters (benchmark-quarters in the lower two panels), and the "portfolio share" rows pool individual positive-weight country or stock positions. The last panel gives each benchmark-quarter equal weight. Country and stock statistics use their respective available observations.

Table A3: Education countries of managers

Country	# Mgrs	Share	Country	# Mgrs	Share
USA	2188	0.7313	ESP	6	0.0020
GBR	580	0.1939	NZL	6	0.0020
CAN	77	0.0257	AUT	5	0.0017
IND	70	0.0234	CHL	5	0.0017
CHN	63	0.0211	POL	4	0.0013
FRA	51	0.0170	PAK	4	0.0013
AUS	44	0.0147	VEN	4	0.0013
DEU	35	0.0117	TUR	3	0.0010
JPN	32	0.0107	ARG	3	0.0010
IRL	26	0.0087	NGA	3	0.0010
HKG	18	0.0060	NOR	3	0.0010
ZAF	16	0.0053	EGY	2	0.0007
ITA	14	0.0047	CZE	2	0.0007
SGP	12	0.0040	GRC	2	0.0007
KOR	11	0.0037	PRT	2	0.0007
CHE	11	0.0037	BGR	1	0.0003
NLD	10	0.0033	ROU	1	0.0003
DNK	10	0.0033	GTM	1	0.0003
SWE	9	0.0030	UKR	1	0.0003
BRA	8	0.0027	THA	1	0.0003
TWN	7	0.0023	EST	1	0.0003
RUS	7	0.0023	LBN	1	0.0003
MEX	7	0.0023	COL	1	0.0003
BEL	7	0.0023	IDN	1	0.0003
ISR	7	0.0023			
Total # of Managers = 2,992					

Notes: Countries are identified by ISO 3166-1 alpha-3 codes (for example, GBR is the United Kingdom, IND India and HKG Hong Kong). Each Share column divides the number of managers educated in the listed country by the 2,992 managers with recorded education countries (49 countries after hand review). The shares sum to more than one because a manager can have education ties to multiple countries.

Table A4: Education-country links across funds

Country	# Funds	Share	Country	# Funds	Share
USA	1492	0.898	AUT	15	0.009
GBR	639	0.384	MEX	14	0.008
IND	188	0.113	POL	14	0.008
CAN	173	0.104	BRA	12	0.007
CHN	133	0.080	VEN	8	0.005
FRA	129	0.078	NGA	8	0.005
AUS	88	0.053	CZE	8	0.005
JPN	67	0.040	PRT	8	0.005
ITA	51	0.031	CHL	7	0.004
IRL	51	0.031	EGY	7	0.004
HKG	46	0.028	ARG	6	0.004
DEU	44	0.026	GRC	6	0.004
SGP	41	0.025	PAK	6	0.004
ZAF	39	0.023	TUR	6	0.004
SWE	38	0.023	ROU	6	0.004
KOR	32	0.019	NOR	6	0.004
ISR	32	0.019	IDN	3	0.002
NLD	32	0.019	THA	2	0.001
CHE	29	0.017	LBN	2	0.001
NZL	28	0.017	GTM	2	0.001
DNK	28	0.017	UKR	2	0.001
ESP	23	0.014	EST	1	0.001
RUS	20	0.012	COL	1	0.001
TWN	19	0.011	BGR	1	0.001
BEL	16	0.010			
Total # of Funds = 1,662					

Notes: Countries are identified by ISO 3166-1 alpha-3 codes, as in Table A3. Each Share column reports the share of funds with at least one manager educated in the listed country: the number of such funds divided by the 1,662 funds with at least one manager whose education country is recorded. This descriptive denominator includes funds with a recorded education link and does not restrict any regression sample. The shares sum to more than one because a fund can have education ties to multiple countries through its managers.

Table A5: The increasing foreign-educated appointment share

Dependent variable	Foreign appointment share	Foreign appointment share
Units	Percentage points	
	(1)	(2)
Specification	Team traits	Without team traits
Calendar years from 2012	0.60***	0.57***
	(0.18)	(0.17)
Log pre-hire destination team size	0.78	1.03
	(1.43)	(1.44)
Unknown education among appointees	−0.78	−0.18
	(5.08)	(5.01)
Female share (known gender)	4.99	—
	(4.03)	
Unknown-gender share	−5.10	—
	(6.76)	
CFA share (observed biographies)	1.06	—
	(3.16)	
Unknown-biography share	5.37	—
	(8.01)	
Mean log(1 + experience)	−0.69	—
	(1.75)	
Fund FE	Yes	Yes
Appointment-year FE	No	No
Observations	4,072	4,072
Fund clusters	1,218	1,218

Notes: The dependent variable is the foreign-educated percentage of known-education appointments. Time is measured in single calendar years from 2012. Both columns retain pre-appointment team size and unknown education among appointees. Column 1 adds five observed team traits; column 2 omits those traits on the identical sample. Appointment shares are weighted by the number of known-education appointments. Standard errors, clustered by economic fund, appear in parentheses below coefficients. *, ** and *** denote two-sided significance at 10%, 5% and 1%.

Table A6: Pre-appointment country relevance and net fund returns

Dependent variable	Factor-adjusted net NAV return	
Units	Annualized bps	
	(1)	(2)
Specification	Baseline controls	Add country shares
Pre-appointment country relevance	−463.18	−688.30***
	(295.12)	(244.59)
Foreign-educated share	32.55	11.82
	(48.28)	(63.37)
Foreign share with unknown relevance	−147.74**	−224.94***
	(58.44)	(61.37)
Log team size	−87.43	−78.48
	(59.12)	(59.75)
Unknown-education share	−106.00	−84.24
	(68.38)	(68.57)
Log lagged assets (USD millions)	−21.75**	−22.49**
	(9.83)	(9.80)
Assets spline basis 1	−101.28***	−105.86***
	(25.89)	(25.73)
Assets spline basis 2	571.88***	602.09***
	(157.85)	(155.38)
Team-size spline basis 1	−80.16	−86.49
	(137.49)	(139.47)
Team-size spline basis 2	688.02	708.37
	(591.04)	(601.64)
Female share (known gender)	−56.00	−58.83
	(38.09)	(38.36)
Unknown-gender share	−72.03	−208.71**
	(91.37)	(96.61)
CFA share (observed biographies)	6.87	4.11
	(31.42)	(31.69)
Unknown-biography share	209.45*	350.28***
	(107.00)	(108.49)
Mean log(1 + experience)	−29.37*	−32.39*
	(16.59)	(16.89)
Prior-year relative NAV return (pp)	−4.63***	−4.83***
	(1.45)	(1.45)
Log(1 + prior relative volatility)	62.83**	60.31**
	(30.99)	(30.65)
Prior-quarter flow (pp)	1.55×10^{-6}	1.57×10^{-6}
	(2.25×10^{-6})	(2.27×10^{-6})
Missing prior return history	99.70	93.63
	(62.75)	(62.37)
Missing prior-quarter flow	−41.63	−41.04
	(55.61)	(55.58)
Lagged expense ratio (bps)	−0.41	−0.45
	(0.33)	(0.33)
Lagged log turnover	85.11**	88.08**
	(35.28)	(34.70)
Top-ten equity concentration	65.36	125.57
	(137.46)	(136.30)
Missing lagged expense ratio	−111.35**	−117.46**
	(53.86)	(53.49)
Missing lagged turnover	73.17**	74.54**
	(36.67)	(36.03)
Missing equity concentration	23.62	36.41
	(42.02)	(41.86)
Education-country composition controls	No	Yes
Fund FE	Yes	Yes
Benchmark × month FE	Yes	Yes
Observations	206,847	206,847
Fund clusters	1,418	1,418

Notes: Both columns add pre-appointment country relevance and the foreign share with unknown relevance to Table 2, column 1. Column 2 also controls for education-country shares. All degree levels enter; definitions appear in Appendix B.1. The relevance coefficient is conditional on foreign-educated share. The foreign-share slopes describe zero measured relevance. Outcomes, sample, baseline controls and equal fund-month weights follow Table 2, column 1. Fund-clustered standard errors appear in parentheses below coefficients. *, ** and *** denote two-sided significance at 10%, 5% and 1%.

Table A7: Portfolio controls and return tails

Dependent variable	Factor-adjusted net NAV return		
Units	Annualized bps		
	(1)	(2)	(3)
Specification	Style controls	Portfolio controls	Winsorized
Foreign-educated share	-65.55^{*}	-68.72^{*}	-68.66^{**}
	(36.51)	(35.44)	(33.08)
Log team size	−89.65	−87.40	-89.27^{*}
	(59.49)	(59.07)	(54.09)
Unknown-education share	−110.43	−106.10	−97.95
	(71.04)	(68.58)	(67.21)
Log lagged assets (USD millions)	-20.96^{**}	-21.48^{**}	-21.04^{**}
	(9.87)	(9.87)	(9.39)
Assets spline basis 1	-100.14^{***}	-97.97^{***}	-92.84^{***}
	(25.91)	(26.13)	(24.74)
Assets spline basis 2	555.51^{***}	547.88^{***}	506.05^{***}
	(158.18)	(158.47)	(149.87)
Team-size spline basis 1	−79.80	−79.90	−68.71
	(138.05)	(137.39)	(129.38)
Team-size spline basis 2	694.90	698.49	647.42
	(592.19)	(589.83)	(560.66)
Female share (known gender)	−49.50	−50.55	−43.03
	(38.37)	(38.53)	(37.14)
Unknown-gender share	−94.23	−88.81	−84.47
	(92.32)	(92.95)	(88.79)
CFA share (observed biographies)	7.61	7.26	3.59
	(31.38)	(31.45)	(30.19)
Unknown-biography share	228.55^{**}	226.85^{**}	213.12^{**}
	(107.30)	(107.68)	(102.99)
Mean log(1 + experience)	-35.24^{**}	-33.29^{**}	-34.78^{**}
	(16.95)	(16.76)	(15.21)
Prior-year relative NAV return (pp)	-4.45^{***}	-4.63^{***}	-4.20^{***}
	(1.54)	(1.47)	(1.37)
Log(1 + prior rel. volatility)	62.66^{**}	60.26^{*}	54.03^{*}
	(31.36)	(31.10)	(27.87)
Prior-quarter flow (pp)	1.57×10^{-6}	1.49×10^{-6}	1.65×10^{-6}
	(2.24×10^{-6})	(2.25×10^{-6})	(2.24×10^{-6})
Missing prior return history	73.20	71.98	79.14
	(56.93)	(59.88)	(57.92)
Missing prior-quarter flow	−44.21	−46.10	−37.84
	(55.44)	(55.55)	(53.42)
Holdings size tilt	−15.74	—	—
	(65.56)		
Holdings momentum tilt	−23.54	—	—
	(38.03)		
Missing size tilt	249.20	—	—
	(371.87)		
Missing momentum tilt	−292.73	—	—
	(346.07)		
Top-ten equity concentration	—	76.34	116.12
		(138.37)	(124.50)
Lagged log turnover	—	82.80^{**}	70.81^{**}
		(35.43)	(33.73)
Missing equity concentration	—	26.52	36.72
		(42.11)	(39.17)
Missing lagged turnover	—	40.67	57.81^{*}
		(31.75)	(33.67)
Lagged expense ratio (bps)	—	—	−0.39
			(0.32)
Missing lagged expense ratio	—	—	-95.70^{*}
			(51.41)
Fund FE	Yes	Yes	Yes
Benchmark × month FE	Yes	Yes	Yes
Observations	206,847	206,847	206,847
Fund clusters	1,418	1,418	1,418

Notes: All columns use thirteen-factor-adjusted NAV. Columns 1–2 add style or concentration and turnover; column 3 adds expenses and 1st/99th-percentile benchmark-month winsorization. Table 2 uses uncapped NAV. All columns retain missing indicators, raw prior flow and equal weights. Fund-clustered standard errors appear in parentheses below coefficients. *, ** and *** denote two-sided significance at 10%, 5% and 1%.

Table A8: Timing and coefficient stability of the fund-return gap

Dependent variable	Factor-adjusted net NAV return		
Units	Annualized bps		
	(1)	(2)	(3)
Specification	Lead sample	With future team	Unknown share only
Foreign-educated share	-74.53^{**}	-72.28^{*}	-58.61^{*}
	(37.81)	(43.87)	(35.54)
Future foreign-educated share	—	−2.60	—
		(37.28)	
Future unknown-education fraction	—	−25.23	—
		(61.67)	
Log team size	−77.10	−77.42	—
	(60.76)	(60.79)	
Unknown-education share	-134.59^{*}	−116.77	−69.42
	(75.46)	(80.49)	(48.05)
Log lagged assets (USD millions)	-21.89^{**}	-21.86^{**}	—
	(9.95)	(9.94)	
Assets spline basis 1	-105.21^{***}	-105.21^{***}	—
	(25.92)	(25.91)	
Assets spline basis 2	587.47^{***}	587.16^{***}	—
	(157.98)	(158.01)	
Team-size spline basis 1	−111.24	−110.75	—
	(142.09)	(142.14)	
Team-size spline basis 2	824.35	822.52	—
	(612.93)	(613.13)	
Female share (known gender)	−56.32	−56.17	—
	(39.90)	(39.93)	
Unknown-gender share	−65.51	−65.84	—
	(101.26)	(101.33)	
CFA share (observed biographies)	1.45	1.44	—
	(33.24)	(33.24)	
Unknown-biography share	226.39^{*}	225.87^{*}	—
	(123.04)	(123.07)	
Mean log(1 + experience)	-39.42^{**}	-39.55^{**}	—
	(17.74)	(17.74)	
Prior-year relative NAV return (pp)	-4.78^{***}	-4.78^{***}	—
	(1.55)	(1.55)	
Log(1 + prior relative volatility)	62.86^{**}	62.82^{**}	—
	(31.62)	(31.61)	
Prior-quarter flow (pp)	1.19×10^{-6}	1.19×10^{-6}	—
	(2.27×10^{-6})	(2.27×10^{-6})	
Missing prior return history	105.34	105.20	—
	(64.36)	(64.37)	
Missing prior-quarter flow	−40.02	−40.01	—
	(56.29)	(56.30)	
Lagged expense ratio (bps)	−0.38	−0.38	—
	(0.43)	(0.43)	
Lagged log turnover	91.39^{**}	91.57^{**}	—
	(35.99)	(35.97)	
Top-ten equity concentration	48.94	49.82	—
	(143.21)	(143.24)	
Missing lagged expense ratio	-110.58^{*}	-110.62^{*}	—
	(62.49)	(62.46)	
Missing lagged turnover	60.13	60.08	—
	(37.94)	(37.93)	
Missing equity concentration	16.94	17.33	—
	(43.68)	(43.71)	
Fund FE	Yes	Yes	Yes
Benchmark × month FE	Yes	Yes	Yes
Observations	193,986	193,986	206,847
Fund clusters	1,418	1,418	1,418

Notes: Risk-adjusted net NAV is in annualized bps. Columns 1–2 require an observed team four quarters ahead; column 2 adds the future team's foreign-educated and unknown-education shares to Equation (3). Column 3 uses only those two current shares on the full sample. Against Table 2, column 1, Oster's δ for a zero coefficient is −20.6 and the bias-adjusted coefficient at Oster's $\delta = 1$ is −72.87 (Oster, 2019), with $R_{\max} = 1.3R^2$. Standard errors clustered by economic fund are in parentheses. *, ** and *** denote two-sided significance at 10%, 5% and 1%.

Table A9: Net performance beyond the initial appointment period

Dependent variable	Factor-adjusted net NAV return	
Units	Annualized bps	
	(1)	(2)
Specification	OLS	WLS
Foreign-educated share: established	−80.91**	−81.82***
	(38.05)	(24.28)
Foreign-educated share: first two years	−37.91	−50.15*
	(40.87)	(27.14)
U.S.-educated share: first two years	−2.56	−12.13
	(21.51)	(14.54)
Log team size	−89.98	−70.02*
	(57.87)	(37.58)
Unknown-education share	−91.04	−123.01***
	(68.25)	(43.69)
Log lagged assets (USD millions)	−16.14*	−12.65*
	(9.44)	(7.33)
Assets spline basis 1	−101.79***	−63.72***
	(26.09)	(18.25)
Assets spline basis 2	548.67***	321.81***
	(158.23)	(106.71)
Team-size spline basis 1	−71.06	−25.95
	(134.60)	(87.51)
Team-size spline basis 2	648.80	388.29
	(578.47)	(371.78)
Female share (known gender)	−51.66	−38.00
	(38.38)	(27.27)
Unknown-gender share	−95.04	−1.06
	(91.55)	(52.86)
CFA share (observed biographies)	9.43	32.91
	(31.25)	(20.91)
Unknown-biography share	230.43**	81.34
	(106.28)	(66.35)
Mean log(1 + experience)	−25.58	−33.77***
	(17.42)	(11.53)
Fund FE	Yes	Yes
Benchmark × month FE	Yes	Yes
Observations	206,847	206,847
Fund clusters	1,418	1,418

Notes: The net NAV model separates managers in their first eight quarters at the fund from established managers; established U.S.-educated managers are the omitted category. Outcomes use the predetermined risk adjustment and reduced controls, in annualized basis points. OLS gives fund-months equal weight; WLS weights by inverse fund residual variance. The first-two-years foreign-minus-U.S. contrasts, the recent foreign-educated coefficient minus the recent U.S.-educated coefficient, are −35.35 (SE 42.14) and −38.02 (27.52), using the joint covariance. Standard errors, clustered by economic fund, appear in parentheses below coefficients. *, ** and *** denote two-sided significance at 10%, 5% and 1%.

Table A10: Conviction portfolios: alternative constructions

Dependent variable	Portfolio excess return							
Units	Annual alpha in bps; factor loadings dimensionless							
Specification	(1) Benchmark reference	(2) Two-year reference	(3) Ex-U.K.	(4) Undergrad.	(5) Momentum	(6) Momentum, no benchmark	(7) Benchmark only	(8) Net costs
Alpha (annualized bps)	275.4*	318.3**	574.1***	492.8***	380.5***	369.8***	379.9***	341.4***
	(147.5)	(132.1)	(148.0)	(134.3)	(136.8)	(136.8)	(124.8)	(124.8)
Global ex-U.S. benchmark excess return	0.101	0.158	0.095	0.182*	0.146	—	1.004***	0.184*
	(0.117)	(0.105)	(0.117)	(0.106)	(0.107)		(0.021)	(0.099)
Developed ex-U.S. market	0.790***	0.775***	0.781***	0.752***	0.780***	0.902***	—	0.749***
	(0.107)	(0.096)	(0.107)	(0.097)	(0.098)	(0.043)		(0.090)
Developed ex-U.S. size	−0.079	−0.065	0.020	−0.001	−0.015	−0.015	—	0.007
	(0.066)	(0.059)	(0.066)	(0.060)	(0.062)	(0.062)		(0.055)
Developed ex-U.S. value	−0.051	−0.028	−0.100	−0.092	−0.001	0.012	—	−0.015
	(0.076)	(0.068)	(0.076)	(0.069)	(0.072)	(0.071)		(0.064)
Developed ex-U.S. investment	−0.104	0.070	0.021	0.087	0.020	0.017	—	−0.073
	(0.103)	(0.092)	(0.104)	(0.094)	(0.096)	(0.097)		(0.087)
Developed ex-U.S. profitability	−0.125	0.052	−0.156	−0.092	0.011	0.025	—	−0.024
	(0.102)	(0.092)	(0.103)	(0.093)	(0.093)	(0.093)		(0.087)
Emerging-market market	0.176***	0.101**	0.162***	0.101**	0.104**	0.128***	—	0.108***
	(0.048)	(0.043)	(0.048)	(0.044)	(0.044)	(0.040)		(0.041)
Emerging-market size	0.044	0.009	−0.056	0.014	−0.004	−0.014	—	0.015
	(0.071)	(0.064)	(0.071)	(0.065)	(0.065)	(0.064)		(0.060)
Emerging-market value	−0.113	−0.029	0.002	−0.034	−0.012	−0.013	—	−0.069
	(0.073)	(0.065)	(0.073)	(0.066)	(0.066)	(0.066)		(0.062)
Emerging-market investment	0.203**	−0.027	−0.080	−0.015	−0.046	−0.048	—	0.082
	(0.085)	(0.076)	(0.086)	(0.078)	(0.078)	(0.078)		(0.072)
Emerging-market profitability	0.125	−0.081	−0.100	0.040	−0.076	−0.080	—	0.022
	(0.094)	(0.084)	(0.094)	(0.085)	(0.087)	(0.087)		(0.079)
Developed ex-U.S. momentum	—	—	—	—	0.001	0.007	—	—
					(0.043)	(0.043)		
Emerging-market momentum	—	—	—	—	0.025	0.021	—	—
					(0.050)	(0.050)		
Observations	300	300	300	300	300	300	300	300

Notes: Each column reports one top-portfolio factor regression. Annual alpha is $12 \times 10^4 \alpha_p$ basis points; loadings are dimensionless. Column 1 replaces the capitalization-share reference in Equation (7) with the assigned benchmark's weights, renormalized over the stocks the fund holds in the country; column 2 widens the reference set to all stocks the fund held in the country over the preceding eight quarters. Columns 3 and 4 exclude U.K. equities and use undergraduate links only; columns 5 and 6 add developed and emerging momentum factors, with and without the common Global ex-U.S. benchmark; columns 7 and 8 use only that benchmark or deduct estimated bid-ask trading costs. All use January 2000–December 2024 and the baseline portfolio-weighting procedure. Portfolio standard errors are conventional OLS; direct Wald tests use the robust stacked covariance. Standard errors appear in parentheses below coefficients. *, ** and *** denote two-sided significance at 10%, 5% and 1%. Top-versus-ordinary Wald p-values for the eight columns are 0.251, 0.057, 0.007, < 0.001, 0.025, 0.025, 0.024 and 0.003, using the robust stacked covariance.

Table A11: Actual equity-capital contributions of education-country holdings

Dependent variable	Excess-return contribution		
Units	Annual alpha in bps; factor loadings dimensionless		
	(1)	(2)	(3)
Specification	All local	Top local	Ordinary local
Alpha (annualized bps)	−9.8	0.3	−10.1*
	(10.8)	(6.1)	(5.8)
Global ex-U.S. benchmark excess return	0.015*	0.008	0.007
	(0.009)	(0.005)	(0.005)
Developed ex-U.S. market	0.078***	0.035***	0.043***
	(0.008)	(0.004)	(0.004)
Developed ex-U.S. size	−0.018***	−0.005*	−0.013***
	(0.005)	(0.003)	(0.003)
Developed ex-U.S. value	0.021***	0.008***	0.013***
	(0.006)	(0.003)	(0.003)
Developed ex-U.S. investment	−0.013*	−0.003	−0.010**
	(0.008)	(0.004)	(0.004)
Developed ex-U.S. profitability	0.029***	0.013***	0.015***
	(0.007)	(0.004)	(0.004)
Emerging-market market	−0.001	−0.004**	0.003*
	(0.004)	(0.002)	(0.002)
Emerging-market size	0.014***	0.006**	0.008***
	(0.005)	(0.003)	(0.003)
Emerging-market value	0.008	0.000	0.007**
	(0.005)	(0.003)	(0.003)
Emerging-market investment	−0.007	0.000	−0.007**
	(0.006)	(0.004)	(0.003)
Emerging-market profitability	−0.006	−0.003	−0.003
	(0.007)	(0.004)	(0.004)
Observations	300	300	300

Notes: Columns 1–3 report capital-weighted excess contributions per dollar of covered disclosed equity, including U.S. equities. Local capital shares average 8.71% overall, 3.62% top and 5.10% ordinary; raw annual contributions are 57.5, 27.3 and 30.1 bps, respectively. All use 300 months, January 2000–December 2024. Loadings are dimensionless; alphas are annualized bps with conventional OLS standard errors. Actual country weights and changing monthly capital shares differ from the benchmark country mix used for the normalized strategies in Table 6. These holdings outcomes are distinct from NAV. Standard errors appear in parentheses below coefficients. *, ** and *** denote two-sided significance at 10%, 5% and 1%.

Table A12: Local alpha on a common sample under benchmark and actual country weights

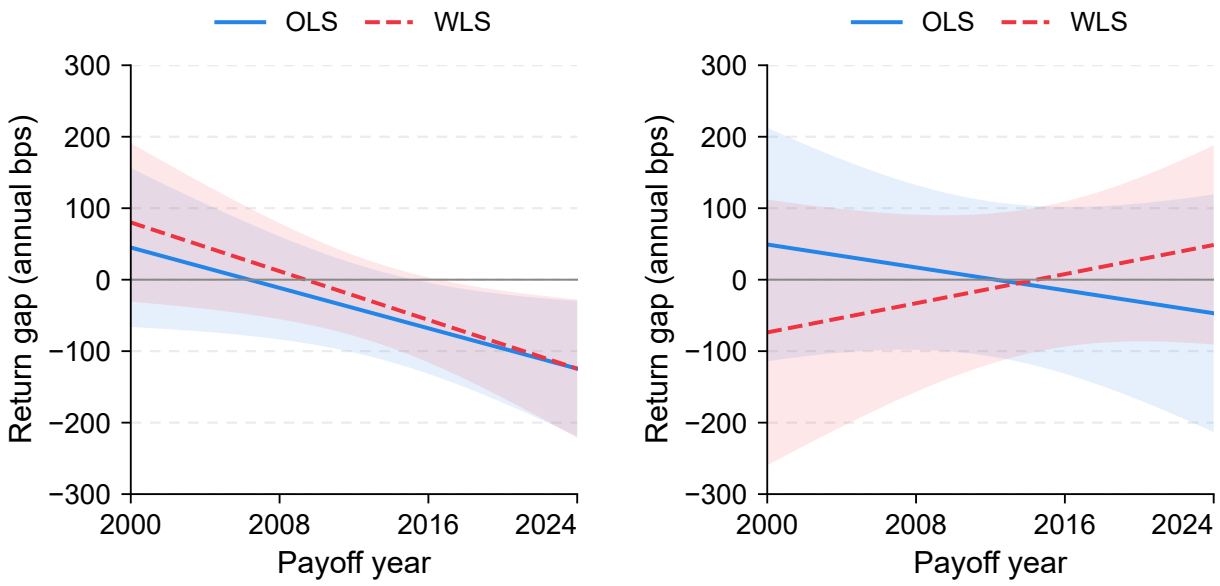
Dependent variable	Portfolio excess return					
Units	Annual alpha in bps; factor loadings dimensionless					
Specification	(1) Benchmark mix: top	(2) Benchmark mix: ordinary	(3) Benchmark mix: combined	(4) Actual weights: top	(5) Actual weights: ordinary	(6) Actual weights: combined
Alpha (annualized bps)	406.7***	3.3	130.8	16.6	−177.5	−103.2
	(134.1)	(92.2)	(94.9)	(159.3)	(109.1)	(119.2)
Global ex-U.S. benchmark excess return	0.155	0.116	0.137*	0.249**	0.117	0.180*
	(0.106)	(0.073)	(0.075)	(0.126)	(0.086)	(0.094)
Developed ex-U.S. market	0.777***	0.731***	0.749***	0.895***	0.858***	0.872***
	(0.097)	(0.067)	(0.069)	(0.115)	(0.079)	(0.086)
Developed ex-U.S. size	−0.007	−0.138***	−0.085**	−0.155**	−0.326***	−0.248***
	(0.060)	(0.041)	(0.042)	(0.071)	(0.048)	(0.053)
Developed ex-U.S. value	−0.008	0.050	0.036	0.170**	0.136**	0.158**
	(0.069)	(0.047)	(0.049)	(0.082)	(0.056)	(0.061)
Developed ex-U.S. investment	0.008	−0.004	0.009	0.018	−0.026	−0.008
	(0.094)	(0.065)	(0.066)	(0.111)	(0.076)	(0.083)
Developed ex-U.S. profitability	0.007	0.033	0.050	0.267**	0.167**	0.220***
	(0.093)	(0.064)	(0.066)	(0.111)	(0.076)	(0.083)
Emerging-market market	0.101**	0.176***	0.138***	−0.113**	0.040	−0.031
	(0.044)	(0.030)	(0.031)	(0.052)	(0.036)	(0.039)
Emerging-market size	0.000	−0.037	−0.024	0.158**	0.121**	0.139**
	(0.065)	(0.045)	(0.046)	(0.077)	(0.053)	(0.058)
Emerging-market value	−0.020	−0.020	−0.016	−0.009	0.071	0.037
	(0.066)	(0.046)	(0.047)	(0.079)	(0.054)	(0.059)
Emerging-market investment	−0.018	−0.075	−0.056	0.057	−0.039	0.008
	(0.078)	(0.053)	(0.055)	(0.092)	(0.063)	(0.069)
Emerging-market profitability	−0.028	−0.006	0.001	0.012	−0.040	−0.014
	(0.085)	(0.059)	(0.060)	(0.101)	(0.069)	(0.076)
Observations	300	300	300	300	300	300

Notes: All portfolios use the same 77,188 fund-country-month cells at 749 fund identifiers and 300 payoff months. Columns 1–3 (benchmark mix) weight each country's holdings by its benchmark country share, as in Table 6; columns 4–6 (actual weights) use the capital funds actually hold in each country. Alpha is annualized; factor loadings are dimensionless monthly slopes with conventional OLS standard errors. Different top/ordinary country mixtures and time-varying capital shares prevent combining their alphas using a single average weight. Standard errors appear in parentheses below coefficients. *, ** and *** denote two-sided significance at 10%, 5% and 1%. The combined actual-weights-minus-benchmark-mix alpha, column 6 minus column 3, is −233.98 bps (SE 85.45, $p = 0.007$), using the joint covariance.

Table A13: Changes in the foreign-education return gap over time

Dependent variable	Adjusted country return	
Units	Annualized bps	
	(1)	(2)
Specification	OLS	WLS
Foreign-educated share × year (γ^T)	-7.06^{**}	-8.52^{**}
	(3.50)	(3.72)
Foreign-educated share (γ)	−39.76	−22.20
	(31.72)	(28.75)
Education-country link × year (β^T)	3.05	13.62^{**}
	(4.98)	(5.40)
Education-country link (β)	40.89	9.61
	(48.78)	(49.80)
Log country stock count	-57.88^{***}	-53.49^{***}
	(11.41)	(11.19)
Log team size	-70.17^{***}	-44.30^{**}
	(21.50)	(18.92)
Unknown-education share	33.08	15.90
	(84.36)	(77.82)
Female share (known gender)	-90.37^{**}	-109.49^{***}
	(44.90)	(37.13)
Unknown-gender share	94.05	134.65
	(87.05)	(82.20)
CFA share (observed biographies)	54.13^{*}	49.53^{*}
	(30.10)	(28.45)
Unknown-biography share	−47.78	−68.59
	(123.27)	(111.68)
Mean log(1 + experience)	-53.72^{***}	-47.69^{***}
	(15.50)	(14.13)
Fund × country FE	Yes	Yes
Country × quarter FE	Yes	Yes
Observations	974,924	974,924
Fund clusters	1,428	1,428

Notes: Equation (17) adds both schooling-link and foreign-share interactions with payoff calendar year minus 2012 to the common panel in Table 5. The first coefficient is the annual change in the outside-country foreign-minus-U.S. return gap; the share coefficient gives that gap in 2012. The 2024 gaps use the entire fitted time profile and the joint coefficient covariance. Shares are in 0–1 units; time interactions are per single year. Column 1 gives countries equal weight within each fund-quarter; column 2 additionally uses inverse fund residual variance. Both columns use fund–country and country–quarter effects, eight displayed controls and economic-fund clustering; the payoff period is 2000–2024. Standard errors appear in parentheses below coefficients. *, ** and *** denote two-sided significance at 10%, 5% and 1%. Fitted 2024 outside gaps are −124.50 (SE 48.28) and −124.50 (49.38); linked-country gaps are −46.96 (84.80) and 48.59 (71.06), in columns 1 and 2. These sums use joint covariance and are plotted in Figure A1.



(a) Outside education countries

(b) Education countries

Figure A1: Education and outside-country selection gaps over time

Notes: Both panels use the joint country regressions in Table A13. Panel A plots $\gamma + \gamma^T t$; Panel B plots $\beta + \gamma + \beta^T t + \gamma^T t$, where t is years from 2012. Each compares an all-foreign-educated team with an all-U.S.-educated team; Panel B additionally requires a current manager's schooling link to the destination. Solid lines are OLS, dashed lines WLS; shaded bands are 95% intervals based on the complete economic-fund-clustered covariance matrix. The local-increment coefficient by itself is not plotted as a team-performance gap.

Table A14: Relatedness and raw country returns outside education countries

Dependent variable	Raw country return		
Units	Annualized bps		
	(1)	(2)	(3)
Specification	Shared border	Official language	Social connectedness
Foreign education × tie (θ)	-345.60^{**}	-194.62^{***}	-81.09^{**}
	(154.53)	(68.89)	(40.85)
Tie (δ)	188.34	132.31^{*}	46.21
	(126.83)	(74.09)	(42.15)
Foreign-educated share (γ)	−27.33	1.41	613.09^{*}
	(33.83)	(34.68)	(318.73)
Missing-tie share (MissingShare)	5.70	34.80	-302.82^{*}
	(47.49)	(48.03)	(168.92)
Log country stock count	-53.07^{***}	-53.25^{***}	-58.83^{***}
	(12.25)	(12.25)	(12.14)
Log team size	-85.29^{***}	-85.81^{***}	-80.88^{***}
	(21.64)	(21.65)	(22.41)
Unknown-education share	−10.45	−9.43	−21.72
	(93.67)	(94.10)	(93.54)
Female share (known gender)	−76.51	−77.01	-90.77^{*}
	(47.08)	(47.02)	(48.93)
Unknown-gender share	37.16	31.78	53.70
	(76.06)	(76.15)	(79.30)
CFA share (observed biographies)	70.73^{**}	70.11^{**}	74.14^{**}
	(31.65)	(31.69)	(31.92)
Unknown-biography share	79.16	85.82	69.45
	(123.40)	(123.29)	(121.99)
Mean log(1 + experience)	-57.50^{***}	-57.83^{***}	-52.22^{***}
	(16.73)	(16.76)	(17.07)
Fund × country FE	Yes	Yes	Yes
Country × quarter FE	Yes	Yes	Yes
Observations	936,076	936,076	876,532
Fund clusters	1,423	1,423	1,421

Notes: The specifications, samples, fixed effects, weights and clustering are those of Table 7; only the dependent variable differs. The raw country return is the return on the fund's holdings in the country over the three months after quarter-end, annualized in basis points, without subtracting the size and past-return matched benchmark of Equation (2). Country-quarter effects still remove the national market return common to all funds holding the country in that quarter. Column 1 is a shared border; column 2 is a common official language; column 3 is log social connectedness from the October 2021 snapshot. The first row is θ, the foreign-minus-U.S. difference in slopes; the tie coefficient δ gives the U.S.-educated slope. Standard errors in parentheses are clustered by economic fund. *, ** and *** denote two-sided significance at 10%, 5% and 1%.

Table A15: Timing of observations contributing to the full-sample NAV association

Timing group	Fund-months	Mean foreign share (%)	Contribution to NAV coefficient (bps)
Before first studied appointment	25,635	41.7	-37.10
Appointment quarter	1,377	56.2	-1.37
First four subsequent quarters	5,133	67.5	-1.65
Years two–five after most recent appointment	13,104	62.2	-6.59
More than five years after most recent appointment	10,377	46.3	-18.21
Funds with no studied appointment	151,221	29.3	-4.59
Full sample	206,847	—	−69.51

Notes: This is an exact descriptive partition of the coefficient in Table 2, not additional regressions. The sample, controls and fixed effects are those of Table 2, column 1. Each contribution sums the product of residual foreign share and residual factor-adjusted NAV in the indicated group, divided by the full-sample sum of squared residual shares. Contributions add to that coefficient; they are neither subgroup return gaps nor hiring effects. Studied appointments are the 460 appointment quarters at 267 funds behind the 511 country cases in Table 9. Timing uses the actual return quarter and resets with another studied appointment. The first studied appointment need not be the first foreign-educated hire; comparison and no-appointment funds can already employ foreign-educated managers. Months are mutually exclusive across rows. Long-horizon observations do not require the original appointee to remain. See Appendix B.11.

Table A16: Foreign education and performance-conditioned monthly flows

Dependent variable	Monthly flow	Monthly flow
Units	Percentage points	
	(1)	(2)
Specification	Full controls	No operating controls
Foreign-educated share	0.207	0.207
	(0.153)	(0.153)
Unknown-education share	−0.458	−0.459
	(0.340)	(0.340)
Log lagged assets	−0.426***	−0.428***
	(0.039)	(0.039)
Log team size	−0.076	−0.078
	(0.087)	(0.087)
Log fund age	−1.991***	−1.986***
	(0.153)	(0.153)
Prior-return rank: bottom segment	5.586***	5.576***
	(0.605)	(0.605)
Prior-return rank: middle segment	1.642***	1.642***
	(0.124)	(0.124)
Prior-return rank: top segment	8.855***	8.863***
	(0.731)	(0.730)
Prior monthly-return volatility (pp)	−0.077*	−0.076*
	(0.043)	(0.043)
Lagged annual expense ratio (bps)	0.0004	—
	(0.0017)	
Log(1 + turnover)	0.104	—
	(0.188)	
Female share (known gender)	−0.361*	−0.363*
	(0.188)	(0.188)
Unknown-gender share	−0.220	−0.227
	(0.381)	(0.382)
CFA share (observed biographies)	−0.051	−0.052
	(0.138)	(0.138)
Unknown-biography share	0.467	0.471
	(0.494)	(0.494)
Mean log(1 + experience)	0.021	0.016
	(0.080)	(0.080)
Fund FE	Yes	Yes
Benchmark × month FE	Yes	Yes
Observations	185,748	185,748
Fund clusters	1,494	1,494

Notes: Equation (18) conditions monthly asset growth net of returns on prior performance ranks, characteristics and team traits. Prior-return rank is on a 0–1 scale, with knots at 0.2 and 0.8; the three spline terms enter the same regression. Column 2 omits expense ratio and turnover on the identical observations; all other regressors and fixed effects are unchanged. The pooled 1st/99th percentile flow caps are −21.435 and 47.589 pp. Fund age is measured from inception. Fund-months have equal weight and standard errors, clustered by economic fund, appear in parentheses below coefficients. *, ** and *** denote two-sided significance at 10%, 5% and 1%.

Table A17: Flows and fees

Dependent variable	Next-quarter flow	Next-quarter flow	Four-quarter flow	Flow after hire	Expense ratio
Units		Flows in pp; expense ratio in annual bps			
	(1)	(2)	(3)	(4)	(5)
Specification	Next quarter	Return sensitivity	Four quarters	After appointment	Expense ratio
Foreign-educated share	−0.29	0.38	−0.26	—	−0.39
	(0.81)	(0.64)	(2.76)		(1.49)
Foreign-educated share × prior-year relative NAV return (pp)	—	−0.04	—	—	—
		(0.06)			
Foreign-educated share of appointees	—	—	—	0.22	—
				(1.97)	
Unknown-education share	−4.36**	−3.14**	−14.41**	—	−3.27
	(1.85)	(1.51)	(6.61)		(3.43)
Log team size	−0.40	−0.26	−2.11	—	−3.73
	(1.40)	(1.18)	(4.65)		(2.35)
Log lagged assets (USD millions)	−8.08***	−4.64***	−22.84***	−21.25***	−0.14
	(0.53)	(0.48)	(1.42)	(2.37)	(0.43)
Assets spline basis 1	3.93***	0.80	0.61	—	−5.04***
	(0.97)	(0.85)	(2.79)		(1.19)
Assets spline basis 2	−8.98*	2.09	28.74*	—	22.27***
	(5.03)	(4.27)	(15.14)		(6.92)
Team-size spline basis 1	−1.86	−1.21	−8.79	—	1.86
	(3.20)	(2.68)	(10.48)		(5.48)
Team-size spline basis 2	10.81	7.75	55.47	—	9.70
	(13.43)	(11.30)	(43.82)		(22.90)
Female share (known gender)	−0.72	−1.38*	−5.29	1.65	−1.14
	(1.07)	(0.83)	(3.55)	(4.28)	(1.61)
Unknown-gender share	−1.11	−1.04	−2.38	−6.94	−1.41
	(1.93)	(1.46)	(6.77)	(12.95)	(2.52)
CFA share (observed biographies)	−0.88	−0.71	−3.65	−3.33	−1.33
	(0.73)	(0.60)	(2.44)	(4.38)	(1.26)
Unknown-biography share	3.12	1.97	8.18	15.67	4.87
	(2.51)	(2.08)	(9.20)	(18.28)	(4.02)
Mean log(1 + experience)	−1.59***	−0.76**	−2.70**	0.24	−1.24**
	(0.40)	(0.33)	(1.27)	(2.41)	(0.61)
Prior-year relative NAV return (pp)	—	0.38***	—	0.68***	—
		(0.03)		(0.17)	
Pre-appointment four-quarter flow (pp)	—	—	—	0.07*	—
				(0.04)	
Log pre-hire destination team size	—	—	—	0.55	—
				(2.62)	
Unknown education among appointees	—	—	—	−1.05	—
				(4.80)	
Fund FE	Yes	Yes	Yes	Yes	Yes
Benchmark × quarter FE	Yes	Yes	Yes	No	Yes
Year FE	No	No	No	Yes	No
Observations	67,343	63,869	62,657	2,973	61,639
Fund clusters	1,418	1,418	1,418	897	1,418

Notes: Flows are net flows as percentages of assets; the final column uses annual expense-ratio basis points. Columns 1–2 explain next-quarter flows, column 3 the next four quarters, and column 4 the four quarters following an appointment conditional on prior flows. Column 2 adds the interaction of foreign-educated share with prior relative performance, in annual NAV percentage points. Team information precedes subsequent flows. All five regressions use equal observation weights and economic-fund clustering. Standard errors appear in parentheses below coefficients. *, ** and *** denote two-sided significance at 10%, 5% and 1%.

Table A18: Prior-record comparisons within origin peers and hiring rounds

Dependent variable	Peer-adjusted return	Prior adjusted return
Units	Annualized basket bps	
	(1)	(2)
Specification	Origin peers	Same hiring round
Foreign-educated appointee	−71.77*	−80.52*
	(40.96)	(41.29)
Log pre-hire destination team size	−13.85	—
	(66.08)	
Pre-hire destination unknown education	422.18**	—
	(206.91)	
Log origin team size	−43.59	69.06
	(55.09)	(91.83)
Origin-team unknown education	161.62	−546.09**
	(330.54)	(260.01)
Log origin country stock count	−68.35	−126.69
	(77.72)	(138.88)
Appointee: top-100 school	54.35	−17.06
	(34.48)	(27.92)
Appointee: school ranked 101–500	134.55**	23.94
	(56.31)	(27.75)
Appointee: female	−18.41	−17.97
	(38.26)	(30.38)
Appointee: unknown gender	43.13	270.17**
	(70.74)	(105.25)
Appointee: CFA	−45.16	−14.23
	(35.06)	(25.84)
Log(1 + manager experience)	22.20	78.88***
	(31.79)	(29.10)
Prior benchmark: China	70.61	—
	(305.31)	
Prior benchmark: Developed Asia-Pacific	−1029.96	—
	(953.70)	
Prior benchmark: Developed Asia-Pacific ex Japan	32.28	—
	(179.00)	
Prior benchmark: Developed Europe	492.95**	—
	(234.47)	
Prior benchmark: Emerging Americas	707.99**	—
	(338.04)	
Prior benchmark: Emerging markets	31.50	—
	(128.52)	
Prior benchmark: Global growth	191.20	—
	(237.58)	
Prior benchmark: Global large cap	127.49	—
	(212.64)	
Prior benchmark: Global	96.97	—
	(164.44)	
Prior benchmark: Global small/mid cap	102.20	—
	(270.14)	
Prior benchmark: Global value	20.29	—
	(220.97)	
Prior benchmark: Global ex-U.S. growth	−66.31	—
	(118.35)	
Prior benchmark: Global ex-U.S. small/mid cap	134.34	—
	(145.04)	
Prior benchmark: Global ex-U.S. value	−23.98	—
	(101.81)	
Prior benchmark: Japan	908.97	—
	(2120.85)	
Hiring-fund FE	Yes	No
Appointment-year FE	Yes	No
Hiring-fund × appointment-quarter FE	No	Yes
Observations	1,321	832
Hiring-fund clusters	300	189

Notes: Both outcomes average four pre-appointment quarters of outside-country adjusted returns. Column 1 subtracts benchmark-country-quarter peers and controls origin benchmarks (Global ex-U.S. omitted). Column 2 compares 832 appointments in 271 common hiring rounds, 126 with both education groups, without peer subtraction or origin-benchmark controls; destination team size and unknown schooling are absorbed. Neither model controls prior NAV. Appointments have equal weight. Hiring-fund-clustered standard errors appear in parentheses below coefficients. *, ** and *** denote two-sided significance at 10%, 5% and 1%.

Table A19: Country coverage and relative departure responses

Dependent variable	Departure	Departure
Units	Probability in pp	
	(1)	(2)
Specification	Other trait slopes	No other trait slopes
Unique coverage × fund return (pp)	0.180***	0.177***
	(0.056)	(0.056)
Duplicate coverage × fund return (pp)	0.006	0.000
	(0.118)	(0.116)
No match × fund return (pp)	0.122**	0.113*
	(0.062)	(0.061)
Unclassified coverage × fund return (pp)	0.051	0.048
	(0.157)	(0.156)
Unique held-country coverage	−0.12	−0.12
	(0.41)	(0.41)
Duplicate held-country coverage	−0.02	−0.01
	(0.78)	(0.78)
Unclassified held-country coverage	−0.83	−0.82
	(0.81)	(0.81)
Log recorded fund spell length	2.70***	2.70***
	(0.28)	(0.28)
Female × fund return (pp)	0.067	—
	(0.056)	
Unknown gender × fund return (pp)	0.012	—
	(0.138)	
CFA × fund return (pp)	−0.051	—
	(0.044)	
Unknown biography × fund return (pp)	−0.111	—
	(0.249)	
Log(1 + experience) × fund return (pp)	0.045	—
	(0.029)	
Log(1 + manager experience)	−3.16***	−3.15***
	(0.41)	(0.41)
Fund × quarter FE	Yes	Yes
Fund × manager FE	Yes	Yes
Observations	154,677	154,677
Fund clusters	1,334	1,334

Notes: Both columns use the same observations and fixed effects. Column 2 omits the five other-trait interactions with performance, retaining spell length and experience. Coverage is measured using prior holdings and incumbent colleagues: unique, duplicated, unmatched or unclassified. Return interactions and their direct differences are percentage points per percentage point of better prior-quarter performance. Downside examples are calculated in the text. These are relative foreign-minus-U.S. exit responses. Managers receive inverse team-size weights, and standard errors, clustered by economic fund, appear in parentheses below coefficients. *, ** and *** denote two-sided significance at 10%, 5% and 1%. In column 1, duplicate minus unique and no-match minus unique return slopes are −0.175 (SE 0.119) and −0.058 (0.076). Jointly testing the duplicate and no-match slopes equal to zero gives $F = 1.96$, $p = 0.141$.

Table A20: Principal variables, units and timing

Variable	Unit	Definition and observation window
Foreign-educated manager share ForeignShare_{jq}	Fraction of known-education managers	Any non-U.S. post-secondary education, including dual U.S./foreign degrees. Quarter-end team used for the following quarter's returns. Measures education location, not nationality or diversity.
Unknown-education share UnknownShare_{jq}	Fraction of all current managers	Managers without identifiable education divided by the recorded team count; entered separately from ForeignShare.
Education-country link	Country indicator or team share	Current managers' recorded education ties to the destination. An outside destination has no such tie.
Factor-adjusted fund return, R^a_{jm}	Annualized fund NAV basis points	Monthly net NAV less the risk-free rate and thirteen predetermined factor-exposure returns. Uses preceding returns and the stated peer fallback; covers the entire fund.
Adjusted country return, R_{jcq}	Annualized basis points	Equation (2): twelve times the mean of the three monthly size/past-return-adjusted returns after quarter-end. Positions drift within the quarter; stocks without returns are dropped. Team and holdings then refresh.
Appointee's prior record, $\bar{R}^{\text{pre}}_h$	Annualized country-basket basis points	Origin-team selection over four full pre-hire payoff quarters. Equal supported-country weights, then averages over quarters and distinct economic origin funds. The main hiring comparison includes all non-U.S. countries and controls origin benchmarks; the $\bar{R}^{\text{peer}}_h$ check uses outside countries only and subtracts benchmark-country-quarter peers. A team record, not personal NAV.
Career performance signal, R^{rel}_{jq}	Quarter-q return in percentage points	Compounded NAV in q relative to other funds sharing the benchmark; predicts careers between the ends of q and $q+1$. Distinct from prior-year controls in the NAV and flow models.
Career outcomes, $\mathbb{1}^{\text{Exit}}_{ijq}$ and $\mathbb{1}^{\text{NewFund}}_{ijq}$	Probability in percentage points	Departure from q to $q+1$, or new recorded appearance at another economic fund in $q+1$. Assignments need not be promotions.
Education-country coverage	Prior-quarter holdings and team education	The specialty-record test uses held countries without a continuing manager educated there. The career test classifies a current manager's held-country links as unique, duplicated by colleagues or unmatched. Missing holdings remain unclassified.
Conviction, I_{jkq}	Within-country active weight	Stock weight less capitalization weight in the same held-country basket. Quartiles precede future returns; portfolio alphas use monthly country-reweighted, factor-adjusted returns.

Notes: The sources, fixed effects, controls and samples of each comparison are described with the corresponding analysis. Panel regressions cluster standard errors by economic fund, except the country-allocation regressions, which cluster by fund identifier; calendar-time conviction portfolios use the time-series covariance specified in Table 6. Education comes from retrospectively collected biographies assigned to dated fund spells.